The equation of a straight line [Cambridge revision]

Gradient, midpoint and length — three formulas that answer almost every coordinate question.

LESSON 53–54 | 2 × 40 MIN | REVISION OF GRADE 8–9 COORDINATES | P1 3.1–3.3

LESSON CLOCK

13' | 21' | 21' | 17' | 8'

The three formulas	13 min
Forms of the equation	21 min
Parallel and perpendicular	21 min
The perpendicular bisector	17 min
Homework	8 min

LEARNING OBJECTIVES

- Find the length, midpoint and gradient of a line segment.
- Write the equation of a line through a point with a given gradient.
- Use the conditions for parallel and perpendicular lines.
- Find the perpendicular bisector of a segment and the foot of a perpendicular.

TEXTBOOK

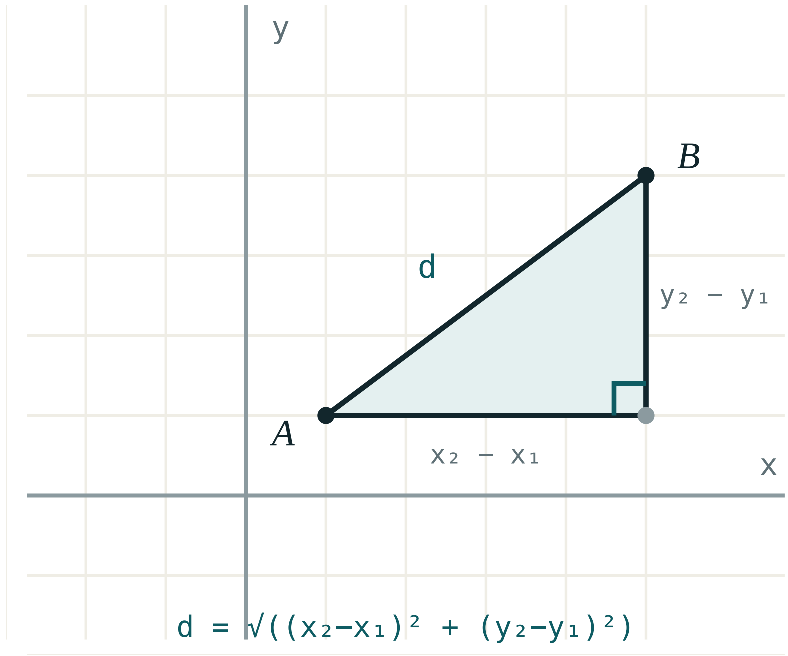
National	Geometry 9, pp. 96–118
Cambridge	Pure Mathematics 1, pp. 44–56
Workbook	Exercise 3A–3C

01 Explanation

The three formulas

For $A(x_1, y_1)$ and $B(x_2, y_2)$:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$



The distance is the hypotenuse of a right triangle with legs Δx and Δy.

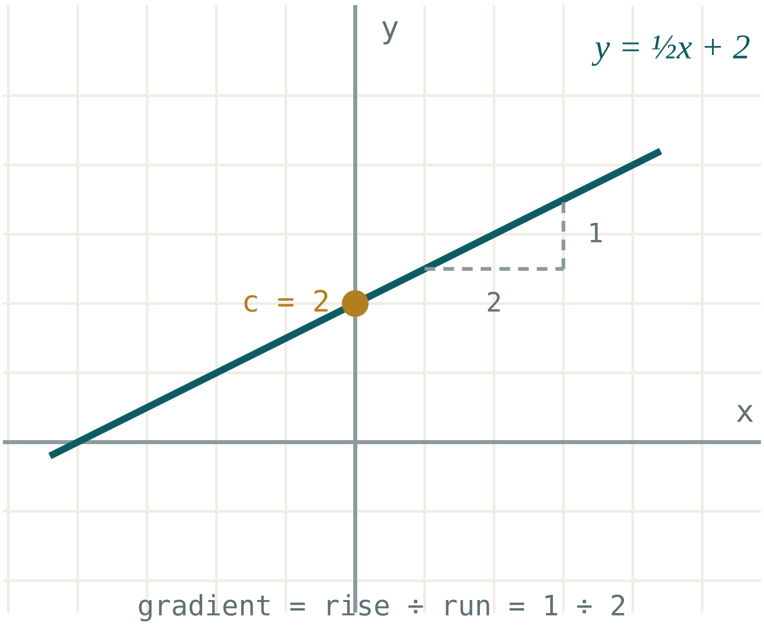
A	B	AB	M	M
$(1, 2)$	$(4, 6)$	5	$(2.5, 4)$	$\frac{4}{3}$
$(-2, 3)$	$(4, -5)$	10	$(1, -1)$	$-\frac{4}{3}$
$(0, 0)$	$(6, 8)$	10	$(3, 4)$	$\frac{4}{3}$

SUBTRACT IN THE SAME ORDER, TWICE

Gradient is $\frac{y_2 - y_1}{x_2 - x_1}$, not $\frac{y_2 - y_1}{x_1 - x_2}$. Reversing one subtraction and not the other flips the sign — and a positive gradient reported as negative loses every mark that follows.

Forms of the equation

FORM	EQUATION	USE IT WHEN
gradient–intercept	$y = mx + c$	the intercept is known or wanted
point–gradient	$y - y_1 = m(x - x_1)$	a point and a gradient are given
two points	$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$	two points are given
general	$ax + by + c = 0$	an answer must be tidy, or the line is vertical



m is the rise over the run; c is where the line crosses Oy.

> **THE VERTICAL LINE HAS NO GRADIENT**
>
> $x = 3$ cannot be written as $y = mx + c$ — its gradient is undefined, not infinite and not zero. A horizontal line, by contrast, has gradient 0 and the perfectly ordinary equation $y = 4$.

Parallel and perpendicular

> parallel: $m_1 = m_2$ perpendicular: $m_1 m_2 = -1$

The second condition says the gradients are **negative reciprocals**: $\frac{2}{3}$ pairs with $-\frac{3}{2}$, and 4 with $-\frac{1}{4}$.

LINE	PARALLEL THROUGH $(1, 5)$	PERPENDICULAR THROUGH $(1, 5)$
$y = 2x + 1$	$y = 2x + 3$	$y = -\frac{1}{2}x + \frac{11}{2}$
$y = -\frac{3}{4}x$	$y = -\frac{3}{4}x + \frac{23}{4}$	$y = \frac{4}{3}x + \frac{11}{3}$
$3x + 2y = 7$	$3x + 2y = 13$	$2x - 3y + 13 = 0$

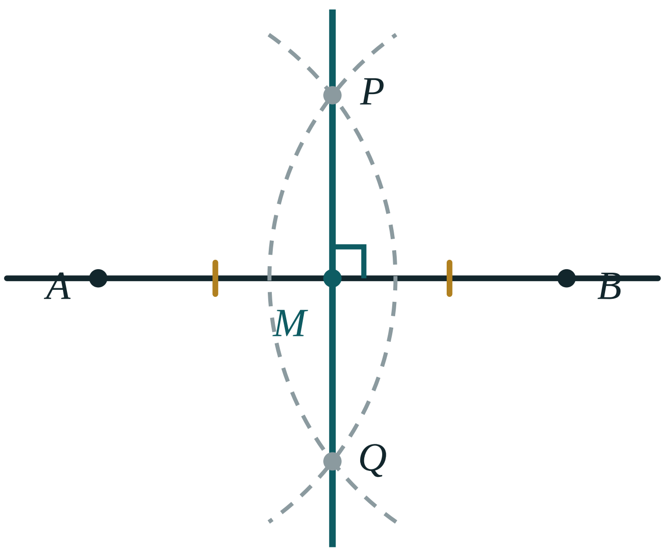
THE SHORTCUT FOR THE GENERAL FORM

A line parallel to $ax + by = c$ is $ax + by = k$; one perpendicular to it is $bx - ay = k$. Swap the coefficients and change one sign — no gradient needed at all.

The perpendicular bisector

The perpendicular bisector of AB is the set of points equidistant from A and B. Two steps build it:

1. the midpoint M of AB — it passes through this;
2. the negative reciprocal of the gradient of AB — this is its gradient.



Every point of the bisector is the same distance from A as from B.

Example. $A(1, 2)$, $B(5, 8)$. Then $M(3, 5)$ and $m\_{AB} = \frac{3}{2}$, so the bisector has gradient $-\frac{2}{3}$:

$$y - 5 = -\frac{2}{3}(x - 3)\text{, that is } 2x + 3y = 21$$

TWO CONDITIONS, BOTH NEEDED

A line through M that is not perpendicular is not the bisector; a perpendicular line that misses M is not either. Check both before writing the answer.

02 Worked examples

EXAMPLE 1 **Find the length, midpoint and gradient of AB for $A(-2, 3)$ and $B(4, -5)$.**

1. $\Delta x = 6, \Delta y = -8$
2. $AB = \sqrt{36 + 64} = 10$
3. $M = (1, -1)$
4. $m = \frac{-8}{6} = -\frac{4}{3}$

ANSWER

$10, (1, -1), -\frac{4}{3}$

EXAMPLE 2 **Find the equation of the line through $(2, -1)$ perpendicular to $y = \frac{3}{4}x + 2$.**

1. $m_1 = \frac{3}{4} \Rightarrow m_2 = -\frac{4}{3}$

   Negative reciprocal.

2. $y + 1 = -\frac{4}{3}(x - 2)$

3. $3y + 3 = -4x + 8$

4. $4x + 3y - 5 = 0$

ANSWER

$4x + 3y - 5 = 0$

EXAMPLE 3 **Find the perpendicular bisector of $A(1, 2)$ and $B(5, 8)$.**

1. $M = (3, 5)$

2. $m\_\{AB\} = \frac{6}{4} = \frac{3}{2}$

3. $m = -\frac{2}{3}$

4. $2x + 3y = 21$

ANSWER

$2x + 3y = 21$

EXAMPLE 4 **Show that $A(1, 1)$, $B(3, 5)$ and $C(6, 11)$ are collinear.**

1. $m_{AB} = \frac{4}{2} = 2$

2. $m_{BC} = \frac{6}{3} = 2$

3. Equal gradients and a common point B.

   So one line.

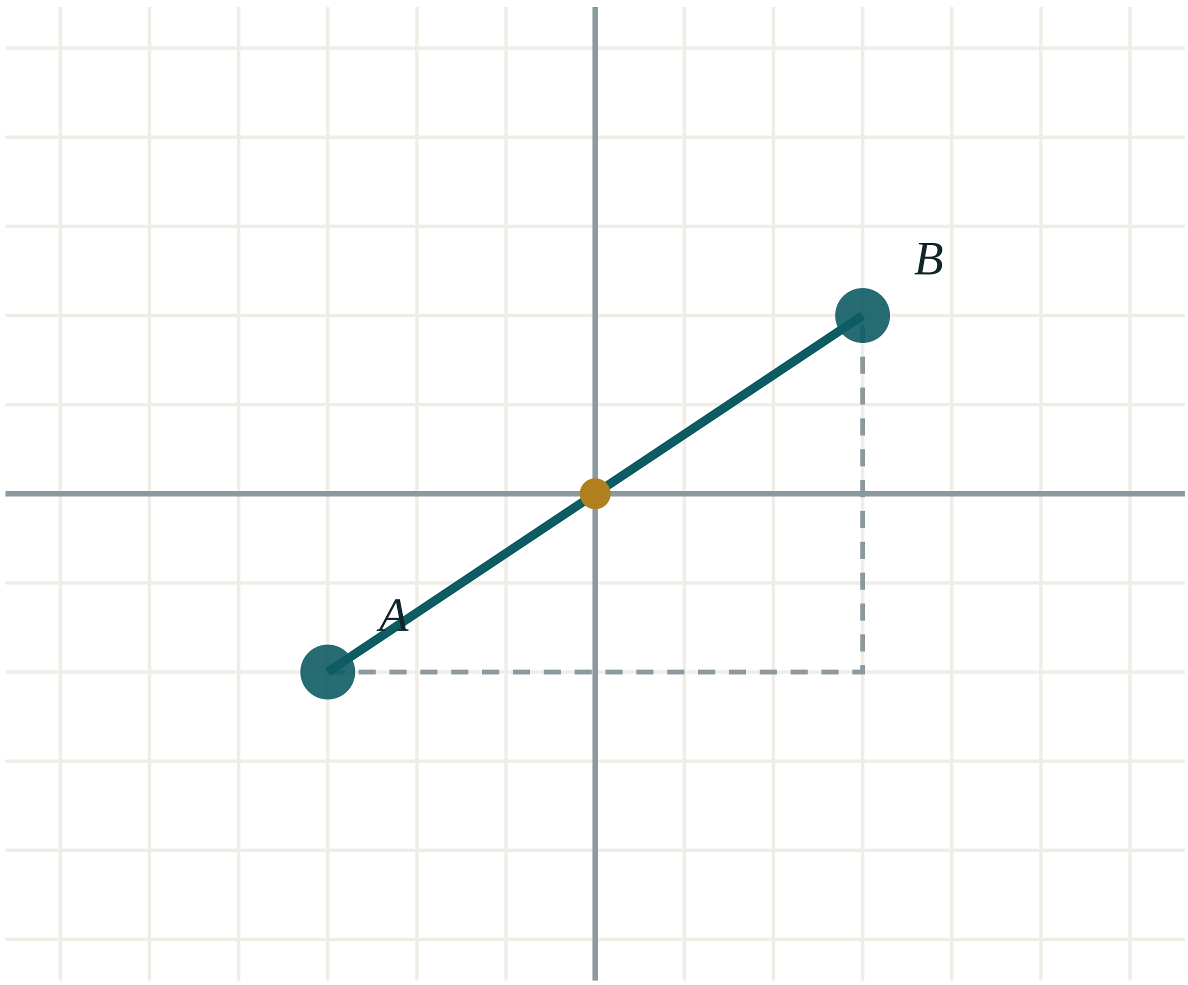
ANSWER

Collinear — both gradients are 2

03 Interactive model

Plot the three points of the collinearity example on squared paper before computing anything.

Gradient, midpoint and length



A **(-3, -2)** | B **(3, 2)** | $x_2 - x_1$ **6** | $y_2 - y_1$ **4** | distance AB **7.21** | midpoint M **(0, 0)**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Coordinate plane	Koordinata tekisligi	Координатная плоскость
Gradient (slope)	Burchak koeffitsiyenti	Угловой коэффициент
Midpoint	O‘rta nuqta	Середина
Distance formula	Masofa formulasi	Формула расстояния
Parallel lines	Parallel to‘g‘ri chiziqlar	Параллельные прямые
Perpendicular lines	Perpendikulyar to‘g‘ri chiziqlar	Перпендикулярные прямые
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Intercept	Kesma	Отрезок на оси
Collinear	Bir to‘g‘ri chiziqda	Коллинеарные
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The gradient of the line through $(1, 2)$ and $(4, 8)$ is:

- 2
- $\frac{1}{2}$
- 6
- 3

Perpendicular gradients satisfy:

- $m_1 = m_2$
- $m_1 m_2 = -1$
- $m_1 + m_2 = 0$
- $m_1 m_2 = 1$

The line $x = 3$ has gradient:

- 0
- 3
- undefined
- ∞

The midpoint of $(-2, 3)$ and $(4, -5)$ is:

- $(1, -1)$
- $(3, -4)$
- $(2, -2)$
- $(1, 4)$

A line parallel to $3x + 2y = 7$ is:

- $2x - 3y = 1$
- $3x + 2y = 13$
- $3x - 2y = 7$
- $2x + 3y = 7$

The perpendicular bisector passes through:

- A
- B
- the midpoint
- the origin

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Length from $(0, 0)$ to $(3, 4)$

   ANSWER 5

2. Midpoint of $(1, 2)$ and $(5, 8)$

   ANSWER $(3, 5)$

3. Gradient through $(1, 2)$ and $(4, 8)$

   ANSWER 2

4. Gradient of $y = -3x + 1$

   ANSWER -3

5. Gradient perpendicular to 2

   ANSWER $-\frac{1}{2}$

6. Equation through $(0, 4)$ with gradient 3

   ANSWER $y = 3x + 4$

7. Gradient of $y = 5$

   ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Length from $(-2, 3)$ to $(4, -5)$

   ANSWER 10

2. Equation through $(2, -1)$ with gradient $\frac{3}{4}$

   ANSWER $3x - 4y - 10 = 0$

3. Equation through $(2, -1)$ perpendicular to $y = \frac{3}{4}x$

   ANSWER $4x + 3y - 5 = 0$

4. Line through $(1, 3)$ and $(5, 11)$

   ANSWER $y = 2x + 1$

5. Perpendicular bisector of $(1, 2)$ and $(5, 8)$

   ANSWER $2x + 3y = 21$

6. Are $(1,1)$, $(3,5)$, $(6,11)$ collinear?

   ANSWER Yes, gradient 2

7. Line parallel to $3x + 2y = 7$ through $(1, 5)$

   ANSWER $3x + 2y = 13$

Hard · stretch and reason

7 PROBLEMS

1. Foot of the perpendicular from $(1, 8)$ to $y = 2x$

   ANSWER $(3.4, 6.8)$

2. Distance from $(1, 8)$ to the line $y = 2x$

   ANSWER $\frac{6}{\sqrt{5}} \approx 2.68$

3. Area of the triangle with vertices $(0,0)$, $(4,0)$, $(2,6)$

   ANSWER 12

4. Find k so that $(1, 2)$, $(3, k)$, $(7, 14)$ are collinear

   ANSWER $k = 6$

5. The vertices $A(0,0)$, $B(4,2)$, $C(2,6)$: show the triangle is right-angled

   ANSWER $m\_\{AB\}\ m\_\{BC\} = \frac{1}{2} \times (-2) = -1$

6. Circumcentre of $(0,0)$, $(6,0)$, $(0,8)$

   ANSWER $(3, 4)$

7. The line $y = mx$ is equidistant from $(0, 4)$ and $(6, 0)$: find m

   ANSWER $m = \frac{2}{3}$

07 Homework

Homework — 5 tasks

Sketch the points before every calculation; a wrong sign shows up in the picture at once.

1. Find the length, midpoint and gradient of AB for $A(-3, 1)$ and $B(5, 7)$.
2. Find the equation of the line through $(3, -2)$ parallel to $2x - 5y = 4$.
3. Find the equation of the perpendicular bisector of $(-1, 4)$ and $(5, -2)$.
4. Show that $A(1, 0)$, $B(4, 6)$ and $C(6, 10)$ are collinear.

5. Find the foot of the perpendicular from $(2, 9)$ to the line $y = 3x + 1$.

The equation of a circle [Cambridge revision]

Centre and radius from the equation, and the equation from the centre and radius.

LESSON 55–56 | 2 × 40 MIN | REVISION OF GRADE 9 | P1 3.4

LESSON CLOCK

12' | 20' | 16' | 20' | 12'

The standard form	12 min
Completing the square	20 min
Inside, on or outside	16 min
Tangents	20 min
Homework	12 min

LEARNING OBJECTIVES

- Write the equation of a circle from its centre and radius.
- Complete the square to read the centre and radius from a general equation.
- Decide whether a point lies inside, on or outside a circle.
- Use the property that the tangent is perpendicular to the radius.

TEXTBOOK

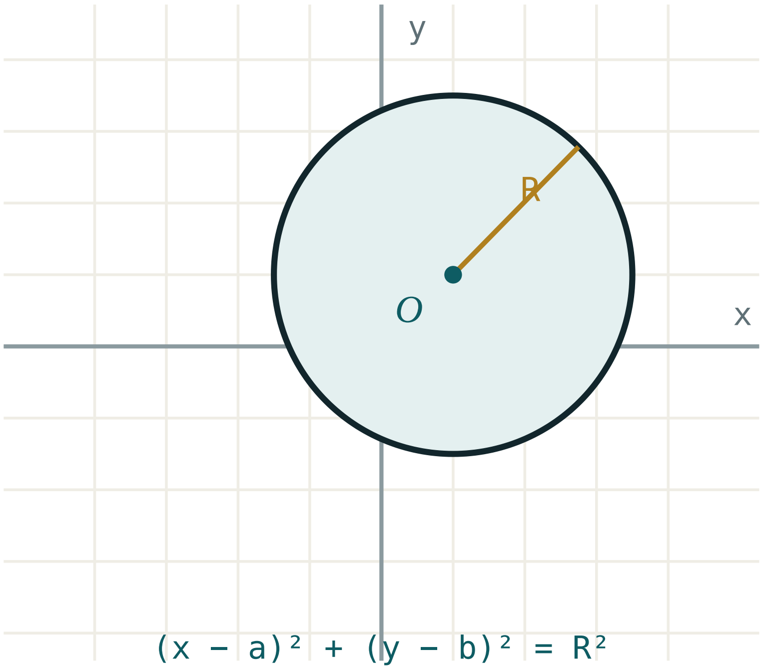
National	Geometry 9, pp. 119–132
Cambridge	Pure Mathematics 1, pp. 57–64
Workbook	Exercise 3D

01 Explanation

The standard form

A circle is the set of points at a fixed distance from a fixed point. Apply the distance formula and square:

$$(x - a)^2 + (y - b)^2 = r^2$$



Centre (a, b), radius r — the distance formula with the square roots removed.

CENTRE	R	EQUATION
$(0, 0)$	5	$x^2 + y^2 = 25$
$(3, -2)$	4	$(x - 3)^2 + (y + 2)^2 = 16$
$(-1, 0)$	$\sqrt{7}$	$(x + 1)^2 + y^2 = 7$

THE SIGNS INSIDE THE BRACKETS ARE REVERSED

A centre at $(3, -2)$ gives $(x - 3)^2$ and $(y + 2)^2$. Reading the centre straight off the brackets without changing the sign is the standard error in this topic.

Completing the square

Expanded, the equation looks unlike a circle:

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Complete the square in x and in y separately to recover the standard form.

Example. $x^2 + y^2 - 6x + 4y - 12 = 0$:

STEP	WORKING
group	$(x^2 - 6x) + (y^2 + 4y) = 12$
complete	$(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12$
tidy	$(x - 3)^2 + (y + 2)^2 = 25$
read	centre $(3, -2)$, radius 5

WHEN IT IS NOT A CIRCLE

If the right-hand side comes out **negative**, no point satisfies the equation; if it is **zero**, the “circle” is the single point at the centre. Both appear in examinations, and both want a sentence, not a radius.

Inside, on or outside

Compute the distance from the point to the centre and compare with r.

$$d < r \text{ inside, } d = r \text{ on, } d > r \text{ outside}$$

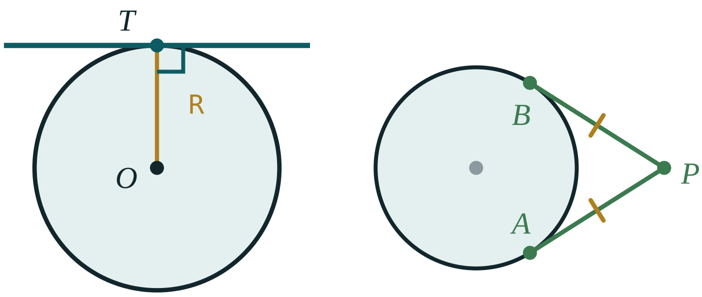
Faster still: substitute the point into $(x - a)^2 + (y - b)^2 - r^2$ and read the sign — negative inside, zero on, positive outside. No square root is needed.

POINT	SUBSTITUTED INTO $(X-3)^2 + (Y+2)^2 - 25$	WHERE
$(3, 3)$	$0 + 25 - 25 = 0$	on
$(5, 0)$	$4 + 4 - 25 = -17$	inside
$(10, 2)$	$49 + 16 - 25 = 40$	outside

Tangents

THE ONE CIRCLE PROPERTY THIS TOPIC NEEDS

The tangent at a point is **perpendicular to the radius** drawn to that point.



Radius and tangent meet at a right angle — the whole of the tangent method.

So the tangent at P on a circle with centre C is found in two steps: compute $m\_\{CP\})$, then use $-\frac{1}{m\_\{CP\}}$ through P.

Example. The tangent to $x^2 + y^2 = 25$ at $(3, 4)$. Here $m\_\{CP\} = \frac{4}{3}$, so the tangent has gradient $-\frac{3}{4}$:

> $y - 4 = -\frac{3}{4}(x - 3)$, that is $3x + 4y = 25$

A PATTERN WORTH REMEMBERING

For the circle $x^2 + y^2 = r^2$ the tangent at (x_0, y_0) is simply $x_0x + y_0y = r^2$. Check: at $(3, 4)$ on $x^2 + y^2 = 25$ it gives $3x + 4y = 25$ at once.

02 Worked examples

EXAMPLE 1 **Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.**

1. $(x^2 - 6x) + (y^2 + 4y) = 12$
2. $(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12$
3. $(x - 3)^2 + (y + 2)^2 = 25$
4. Centre $(3, -2)$, radius 5.

ANSWER

Centre $(3, -2)$, radius 5

EXAMPLE 2 **Find the equation of the circle with $A(1, 2)$ and $B(7, 10)$ as ends of a diameter.**

1. Centre = midpoint = $(4, 6)$.
2. $AB = \sqrt{36 + 64} = 10 \Rightarrow r = 5$
3. $(x - 4)^2 + (y - 6)^2 = 25$

ANSWER

$(x - 4)^2 + (y - 6)^2 = 25$

EXAMPLE 3 **Where does $(6, 1)$ lie relative to $(x - 3)^2 + (y + 2)^2 = 25$?**

1. $(6 - 3)^2 + (1 + 2)^2 = 9 + 9 = 18$
2. $18 < 25$
3. Inside.

   No square root needed.

ANSWER

Inside the circle

EXAMPLE 4 **Find the tangent to $x^2 + y^2 = 25$ at $(-4, 3)$.**

1. $m_{CP} = -\dfrac{3}{4}$
2. $m_{tangent} = \dfrac{4}{3}$
3. $y - 3 = \dfrac{4}{3}(x + 4)$
4. $4x - 3y + 25 = 0$

   Or use $x_0x + y_0y = 25$ directly.

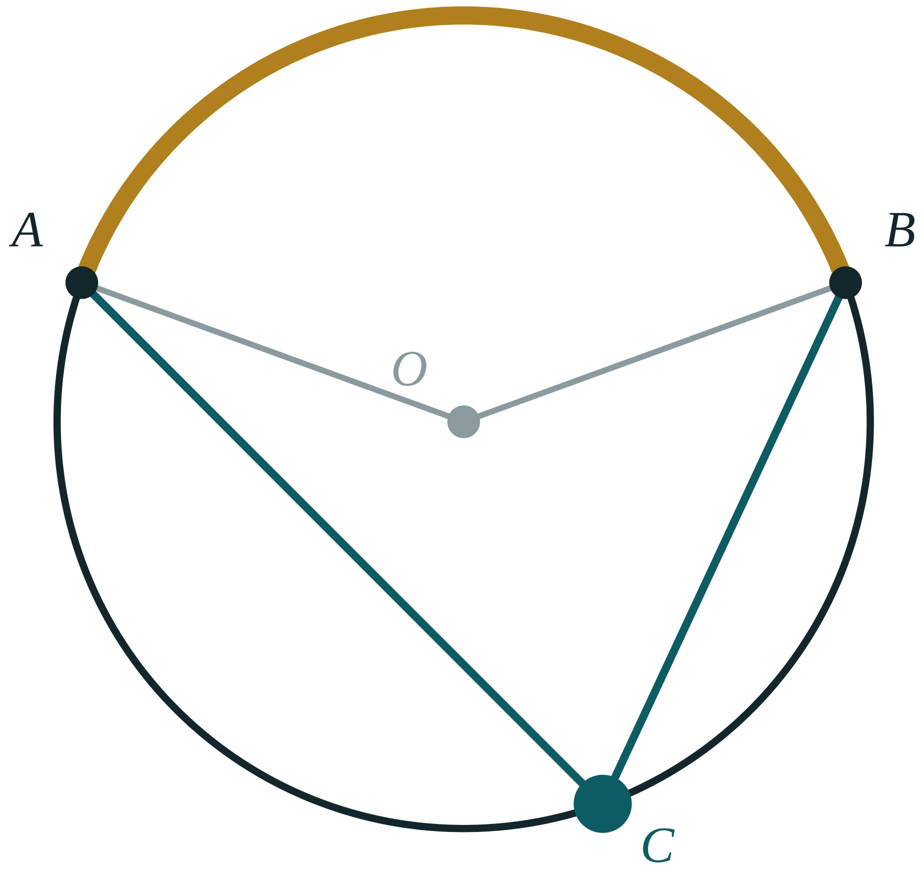
ANSWER

$4x - 3y + 25 = 0$

03 Interactive model

Complete the square on the board twice — once with a positive and once with a negative constant — and let the class say what the second one is.

The circle and its lines



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Circle	Aylana	Окружность
Centre	Markaz	Центр
Radius	Radius	Радиус
Standard form	Kanonik ko‘rinish	Каноническое уравнение
General form	Umumiy ko‘rinish	Общее уравнение
Completing the square	To‘la kvadratga keltirish	Выделение полного квадрата
Tangent	Urinma	Касательная
Chord	Vatar	Хорда
Diameter	Diametr	Диаметр
Concentric	Konsentrik	Концентрические

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The circle $(x - 3)^2 + (y + 2)^2 = 16$ has centre:

- $(3, 2)$
- $(3, -2)$
- $(-3, 2)$
- $(-3, -2)$

Its radius is:

- 16
- 4
- 8
- $\sqrt{16}^{2}$

To find the centre from the general form you:

- factorise
- complete the square
- differentiate
- substitute

If the right side comes out negative, the equation gives:

- a small circle
- no points at all
- a point
- a line

A tangent meets the radius at:

- $45°$
- $90°$
- $60°$
- any angle

The tangent to $x^2 + y^2 = 25$ at $(3, 4)$ is:

- $3x + 4y = 25$
- $4x + 3y = 25$
- $3x - 4y = 25$
- $x + y = 7$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Equation of the circle, centre $(0, 0)$, radius 5

   ANSWER $x^2 + y^2 = 25$

2. Equation, centre $(3, -2)$, radius 4

   ANSWER $(x - 3)^2 + (y + 2)^2 = 16$

3. Centre of $(x + 1)^2 + (y - 5)^2 = 9$

   ANSWER $(-1, 5)$

4. Radius of $(x + 1)^2 + (y - 5)^2 = 9$

   ANSWER 3

5. Radius of $x^2 + y^2 = 20$

   ANSWER $2\sqrt{5}$

6. Does $(3, 4)$ lie on $x^2 + y^2 = 25$?

   ANSWER Yes

7. Centre of $x^2 + y^2 = 49$

   ANSWER $(0, 0)$

Medium · the standard question

7 PROBLEMS

1. Centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$

   ANSWER $(3, -2), 5$

2. Centre and radius of $x^2 + y^2 + 8x - 2y + 8 = 0$

   ANSWER $(-4, 1), 3$

3. Circle on the diameter $(1, 2)$ to $(7, 10)$

   ANSWER $(x - 4)^2 + (y - 6)^2 = 25$

4. Is $(6, 1)$ inside $(x - 3)^2 + (y + 2)^2 = 25$?

   ANSWER Yes — $18 < 25$

5. Tangent to $x^2 + y^2 = 25$ at $(3, 4)$

   ANSWER $3x + 4y = 25$

6. Tangent to $x^2 + y^2 = 25$ at $(-4, 3)$

   ANSWER $4x - 3y + 25 = 0$

7. Circle with centre $(2, 3)$ passing through $(5, 7)$

   ANSWER $(x - 2)^2 + (y - 3)^2 = 25$

Hard · stretch and reason

7 PROBLEMS

1. Show that $x^2 + y^2 - 4x + 6y + 20 = 0$ has no points

   ANSWER $r^2 = -7$

2. Circle through $(0,0)$, $(6,0)$, $(0,8)$

   ANSWER $(x - 3)^2 + (y - 4)^2 = 25$

3. Length of the tangent from $(10, 2)$ to $(x - 3)^2 + (y + 2)^2 = 25$

   ANSWER $\sqrt{40} \approx 6.32$

4. The circles $x^2 + y^2 = 4$ and $(x - 5)^2 + y^2 = 9$: do they touch?

   ANSWER Yes, externally — $5 = 2 + 3$

5. Shortest distance from $(10, 2)$ to that circle

   ANSWER $\sqrt{65} - 5 \approx 3.06$

6. Circle with centre on $y = x$, radius 5, through $(1, 1)$

   ANSWER $(x - a)^2 + (y - a)^2 = 25$ with $a = 1 \pm \dfrac{5}{\sqrt{2}}$

7. Equation of the circle inscribed in the square with vertices $(0,0)$, $(6,0)$, $(6,6)$, $(0,6)$

   ANSWER $(x - 3)^2 + (y - 3)^2 = 9$

07 Homework

Homework — 5 tasks

Show the completing-the-square step in full; the answer alone earns half the marks.

1. Find the centre and radius of $x^2 + y^2 - 10x + 6y + 9 = 0$.
2. Find the equation of the circle with $(-2, 1)$ and $(6, 7)$ as ends of a diameter.
3. Decide whether $(1, 1)$, $(5, -3)$ and $(9, 4)$ lie inside, on or outside $(x - 5)^2 + (y + 3)^2 = 16$.
4. Find the tangent to $x^2 + y^2 = 100$ at $(6, 8)$, and check it with $x_0x + y_0y = r^2$.

5. Explain in two sentences why $x^2 + y^2 + 2x + 2y + 5 = 0$ has no solutions.

Points of intersection of lines and circles [Cambridge revision]

Substitute, solve the quadratic, and let the discriminant say how many points there are.

LESSON 57–58 | 2 × 40 MIN | REVISION OF GRADE 9 | P1 3.5

LESSON CLOCK

12' | 20' | 20' | 16' | 12'

Two lines	12 min
A line and a circle	20 min
The discriminant decides	20 min
The tangent condition	16 min
Homework	12 min

LEARNING OBJECTIVES

- Find where two lines meet by solving their equations simultaneously.
- Find where a line meets a circle by substitution.
- Use the discriminant to count intersections without solving.
- Find the condition for a line to be a tangent to a circle.

TEXTBOOK

National	Geometry 9, pp. 133–146
Cambridge	Pure Mathematics 1, pp. 65–72
Workbook	Exercise 3E

01 Explanation

Two lines

Two lines meet where both equations hold. Solve simultaneously — by substitution when one equation is already $y = \ldots$, by elimination otherwise.

PAIR	RELATION	MEETING POINTS
$y = 2x + 1, y = -x + 7$	different gradients	$(2, 5)$
$y = 2x + 1, y = 2x + 5$	parallel	none
$y = 2x + 1, 4x - 2y + 2 = 0$	the same line	infinitely many

THREE OUTCOMES, AND WHAT EACH LOOKS LIKE ALGEBRAICALLY

A unique solution, no solution (a contradiction such as $0 = 4$), or infinitely many (an identity such as $0 = 0$). The algebra always tells you which before the picture does.

A line and a circle

Make y the subject of the line, substitute into the circle, and a quadratic in x appears. Its roots are the x-coordinates of the meeting points.

Example. $y = x + 1$ and $x^2 + y^2 = 25$:

STEP	WORKING
substitute	$x^2 + (x + 1)^2 = 25$
expand	$2x^2 + 2x - 24 = 0$
simplify	$x^2 + x - 12 = 0$
solve	$x = 3, x = -4$
back-substitute	$(3, 4)$ and $(-4, -3)$

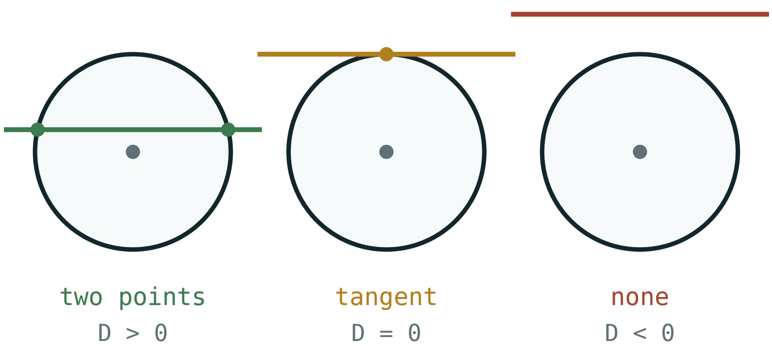
FINISH THE JOB — FIND Y TOO

A point has two coordinates. Stopping at $x = 3, x = -4$ answers half the question and earns half the marks. Substitute back into the **line**, which is easier than the circle.

The discriminant decides

The number of meeting points is the number of roots of that quadratic — so it is decided by $D = b^2 - 4ac$ alone.

$D > 0$ two points $D = 0$ tangent $D < 0$ no points



Three positions, three signs of the discriminant.

This lets you answer "how many times does the line meet the circle?" without solving anything — form the quadratic, compute D, stop.

LINE WITH $X^2 + Y^2 = 25$	QUADRATIC	D	POINTS
$y = x + 1$	$x^2 + x - 12 = 0$	49	2
$y = x + 5\sqrt{2}$	$2x^2 + 10\sqrt{2}x + 25 = 0$	0	1
$y = x + 10$	$2x^2 + 20x + 75 = 0$	-200	0

The tangent condition

A line is a tangent exactly when $D = 0$. Setting $D = 0$ and solving for the unknown in the line gives every tangent of that family.

Example. For which c is $y = x + c$ a tangent to $x^2 + y^2 = 25$?

$$2x^2 + 2cx + c^2 - 25 = 0, D = 4c^2 - 8(c^2 - 25) = 200 - 4c^2$$

So $D = 0$ when $c^2 = 50$, that is $c = \pm 5\sqrt{2}$ — the two tangents parallel to $y = x$, one on each side.

THE GEOMETRIC ROUTE, OFTEN FASTER

A line is a tangent when the distance from the centre to the line equals r:

$$d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} = r$$

For $y = x + c$ and the centre $(0, 0)$: $\frac{|c|}{\sqrt{2}} = 5$, giving $c = \pm 5\sqrt{2}$ in one line.

Chord length. If $d < r$, the half-chord, d and r form a right triangle, so the chord is $2\sqrt{r^2 - d^2}$ — no quadratic needed.

02 Worked examples

EXAMPLE 1 **Find where $y = x + 1$ meets $x^2 + y^2 = 25$.**

1. $x^2 + (x + 1)^2 = 25$
2. $2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0$
3. $x = 3$ or $x = -4$
4. $(3, 4)$ and $(-4, -3)$

   Back into the line.

ANSWER

$(3, 4)$ and $(-4, -3)$

EXAMPLE 2 **How many times does $y = 2x + 8$ meet $x^2 + y^2 = 9$?**

1. $x^2 + (2x + 8)^2 = 9$
2. $5x^2 + 32x + 55 = 0$
3. $D = 1024 - 1100 = -76$
4. Negative — the line misses the circle.

ANSWER

No points of intersection

EXAMPLE 3 **For which c is $y = 2x + c$ a tangent to $x^2 + y^2 = 20$?**

1. Distance from $(0,0)$: $\dfrac{|c|}{\sqrt{5}}$

2. $\dfrac{|c|}{\sqrt{5}} = \sqrt{20} = 2\sqrt{5}$

3. $|c| = 10$

4. $c = \pm 10$

ANSWER

$c = 10$ or $c = -10$

EXAMPLE 4 **Find the length of the chord cut on $x^2 + y^2 = 25$ by $y = 3$.**

1. $d = 3$ from the centre.

2. Half-chord $\sqrt{25 - 9} = 4$

   Right triangle.

3. Chord $= 8$

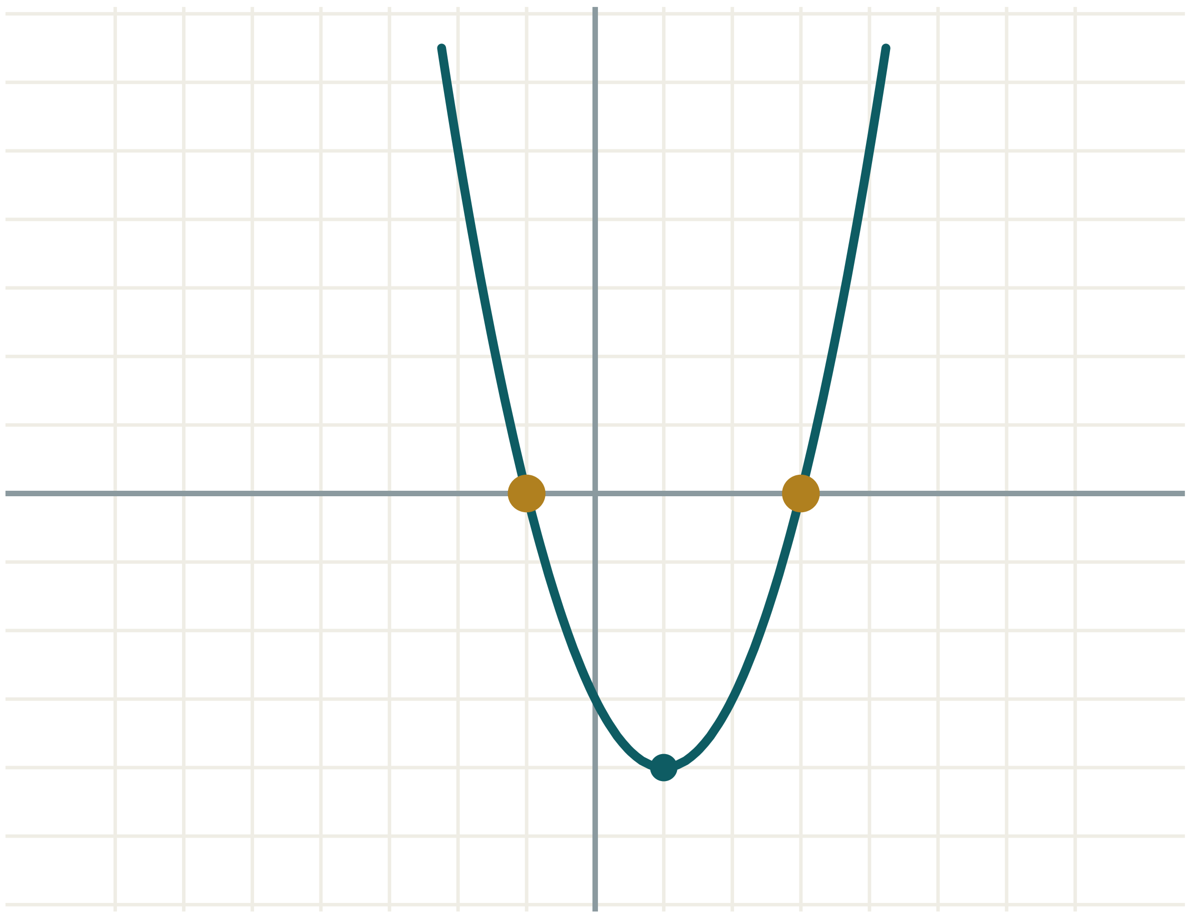
ANSWER

8

03 Interactive model

Draw one circle and slide a ruler across it, stopping at each of the three positions; name the sign of D each time.

The discriminant, and how many points



a **1**	b **-2**	c **-3**	$D = b^2 - 4ac$ **16**	roots **two**	x_1 **-1**	x_2 **3**	$x_1 + x_2$ **2**	$x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Point of intersection	Kesishish nuqtasi	Точка пересечения
Simultaneous equations	Tenglamalar sistemasi	Система уравнений
Substitution	O‘rniga qo‘yish	Подстановка
Discriminant	Diskriminant	Дискриминант
Tangent condition	Urinma sharti	Условие касания
Secant	Kesuvchi	Секущая
Chord length	Vatar uzunligi	Длина хорды
Distance from a point to a line	Nuqtadan chiziqqa masofa	Расстояние от точки до прямой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A line meets a circle in at most:

- one point
- two points
- three
- four

$D = 0$ means the line is:

- a secant
- a tangent
- missing the circle
- a diameter

After finding x you must:

- stop
- find y from the line
- find y from the circle
- square it

Two parallel lines give:

- one point
- no solution
- infinitely many
- two points

A line is a tangent when its distance from the centre is:

- zero
- equal to r
- less than r
- greater than r

The chord at distance d from the centre has length:

- $\sqrt{r^2 - d^2}$
- $2\sqrt{r^2 - d^2}$
- $2d$
- $r - d$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Where do $y = 2x + 1$ and $y = -x + 7$ meet?

   ANSWER $(2, 5)$

2. Where do $y = 3x$ and $y = 3x + 2$ meet?

   ANSWER Nowhere — parallel

3. Does $(3, 4)$ lie on $x^2 + y^2 = 25$?

   ANSWER Yes

4. D of $x^2 + x - 12 = 0$

   ANSWER 49

5. Chord of $x^2 + y^2 = 25$ along $y = 3$

   ANSWER 8

6. Chord of $x^2 + y^2 = 25$ along $y = 0$

   ANSWER 10

7. Distance from $(0,0)$ to $y = x + 4$

   ANSWER $\frac{4}{\sqrt{2}} = 2\sqrt{2}$

Medium · the standard question

7 PROBLEMS

1. Where does $y = x + 1$ meet $x^2 + y^2 = 25$?

   ANSWER $(3, 4), (-4, -3)$

2. How many points has $y = 2x + 8$ with $x^2 + y^2 = 9$?

   ANSWER None — $D = -76$

3. For which c is $y = 2x + c$ a tangent to $x^2 + y^2 = 20$?

   ANSWER $c = \pm 10$

4. For which c is $y = x + c$ a tangent to $x^2 + y^2 = 25$?

   ANSWER $c = \pm 5\sqrt{2}$

5. Where does $x = 4$ meet $x^2 + y^2 = 25$?

   ANSWER $(4, 3), (4, -3)$

6. Chord of $x^2 + y^2 = 100$ along $y = 6$

   ANSWER 16

7. Where does $y = 2x$ meet $(x - 5)^2 + y^2 = 5$?

   ANSWER $(1, 2), (4, 8)$

Hard · stretch and reason

7 PROBLEMS

1. Length of the chord of $x^2 + y^2 = 25$ on $y = x + 1$

   ANSWER $7\sqrt{2} \approx 9.90$

2. The tangents from $(0, 10)$ to $x^2 + y^2 = 25$

   ANSWER $y = \pm\sqrt{3}\,x + 10$

3. For which k does $y = kx$ touch $(x - 5)^2 + y^2 = 9$?

   ANSWER $k = \pm\dfrac{3}{4}$

4. Where do $x^2 + y^2 = 25$ and $(x - 8)^2 + y^2 = 25$ meet?

   ANSWER $(4, 3), (4, -3)$

5. Shortest distance from $(8, 6)$ to $x^2 + y^2 = 25$

   ANSWER 5

6. The line $3x + 4y = 25$ and the circle $x^2 + y^2 = 25$: how many points?

   ANSWER One — it is the tangent at $(3, 4)$

7. For which r does $y = x + 6$ touch $x^2 + y^2 = r^2$?

   ANSWER $r = 3\sqrt{2}$

07 Homework

Homework — 5 tasks

Always give both coordinates of every point you find.

1. Find where $y = 2x - 5$ meets $x^2 + y^2 = 25$.
2. Show that $y = x + 8$ does not meet $x^2 + y^2 = 25$, using the discriminant.
3. Find the values of c for which $y = 3x + c$ is a tangent to $x^2 + y^2 = 40$.
4. Find the length of the chord cut on $x^2 + y^2 = 169$ by the line $y = 5$.

5. Find where the circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$ intersect.

Vectors in the plane and transformations [Cambridge revision]

A vector is a journey, not a place — and every transformation is one rule applied to all of them.

LESSON 59–60 | 2 × 40 MIN | REVISION OF GRADE 8–9 VECTORS | IGX 23.1–23.3

LESSON CLOCK

16' | 20' | 16' | 20' | 8'

Vectors as journeys	16 min
Arithmetic of vectors	20 min
Proving with vectors	16 min
The four transformations	20 min
Homework	8 min

LEARNING OBJECTIVES

- Add, subtract and scale vectors in column form and geometrically.
- Compute the magnitude of a vector and the position vector of a point.
- Use vectors to prove that lines are parallel or that points are collinear.
- Describe fully a translation, reflection, rotation and enlargement.

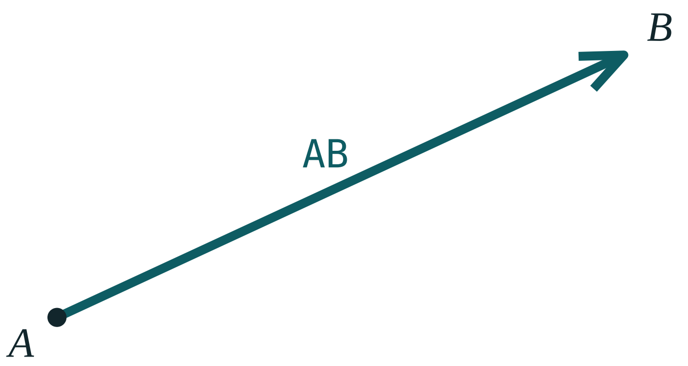
TEXTBOOK

National	Geometry 9, pp. 147–170
Cambridge	Core & Extended, pp. 590–612
Workbook	Exercise 23.1–23.3

01 Explanation

Vectors as journeys

A vector has **magnitude** and **direction** but no position. The arrow from $(1, 1)$ to $(4, 5)$ and the arrow from $(0, 0)$ to $(3, 4)$ are the **same vector** $\begin{pmatrix}3\\4\end{pmatrix}$.



One vector, drawn twice — same length, same direction, different place.

$$\left|\begin{pmatrix} a \\ b \end{pmatrix}\right| = \sqrt{a^2 + b^2}$$

THE POSITION VECTOR FIXES IT IN PLACE

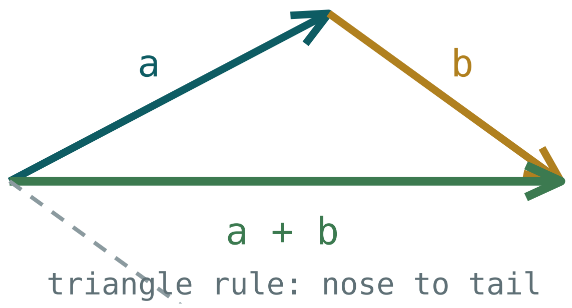
The position vector of A is OA, drawn from the origin. Then for any two points

$$AB = OB - OA \text{ (“to minus from”)}$$

Getting this the wrong way round reverses every vector in the question, so learn it as a sentence.

Arithmetic of vectors

OPERATION	IN COLUMNS	GEOMETRICALLY
add	$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$	nose to tail
subtract	$\begin{pmatrix} a \\ b \end{pmatrix} - \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a-c \\ b-d \end{pmatrix}$	add the reverse
scale	$k\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$	same line, k times as long



Nose to tail — the resultant closes the triangle.

PARALLEL MEANS A SCALAR MULTIPLE

u and v are parallel exactly when $v = ku$ for some number k. So $\begin{pmatrix}3\\4\end{pmatrix}$ and $\begin{pmatrix}9\\12\end{pmatrix}$ are parallel ($k = 3$); $\begin{pmatrix}3\\4\end{pmatrix}$ and $\begin{pmatrix}4\\3\end{pmatrix}$ are not.

Unit vector. Divide by the magnitude: the unit vector along $\begin{pmatrix}3\\4\end{pmatrix}$ is $\frac{1}{5}\begin{pmatrix}3\\4\end{pmatrix}$.

Proving with vectors

Two standard proofs cover most questions.

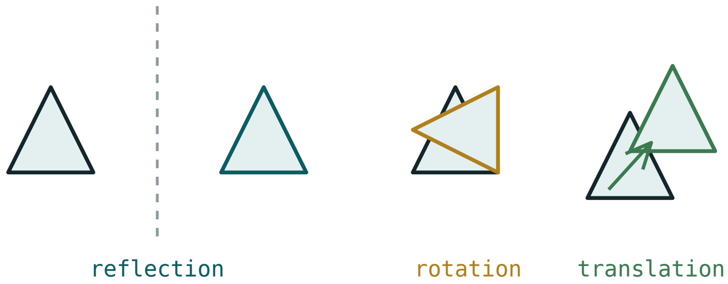
TO PROVE	SHOW
AB is parallel to CD	$AB = k{\cdot}CD$
A, B, C are collinear	$AB = k{\cdot}BC$
$ABCD$ is a parallelogram	$AB = DC$
M is the midpoint of AB	$OM = \frac{1}{2}(OA + OB)$

The collinearity test differs from the parallel test only in that the two vectors share a **point**. Parallel plus a common point means one line — that is the whole argument, and it should be written in the answer.

A VECTOR PROOF NEEDS A CONCLUDING SENTENCE

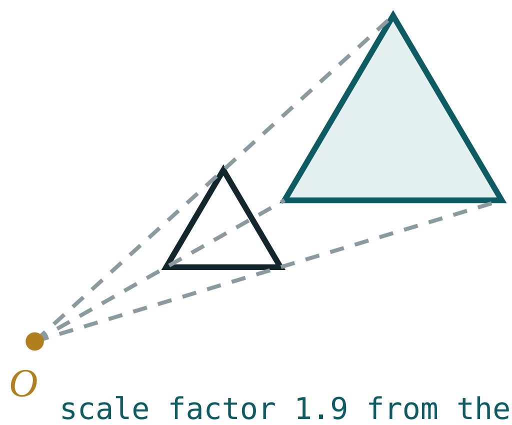
> "$AB = 2 \cdot BC$, and B is common to both, so A, B and C are collinear." Without the second half the algebra proves only parallelism.

The four transformations



Reflection, rotation and translation — each carries the shape to a congruent copy.

TRANSFORMATION	FULLY DESCRIBED BY	PRESERVES
translation	a column vector	size, shape, direction
reflection	the mirror line	size and shape; reverses sense
rotation	centre, angle and sense	size and shape
enlargement	centre and scale factor k	shape only



An enlargement from a centre — lengths scale by k, areas by k^2.

"FULLY DESCRIBE" IS A MARK SCHEME INSTRUCTION

A rotation needs **three** pieces of information; giving two earns part marks. An enlargement needs the centre as well as k. And a negative k puts the image on the other side of the centre, inverted.

Under an enlargement of factor k: lengths $\times\ k$, areas $\times\ k^2$. Under the other three, lengths and areas are unchanged — they are **congruences**.

02 Worked examples

EXAMPLE 1 **With $A(1, 2)$ and $B(5, 9)$, find AB and $|AB|$.**

1. $AB = OB - OA$

   "to minus from".

2. $= \begin{pmatrix}5\\9\end{pmatrix} - \begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}4\\7\end{pmatrix}$

3. $|AB| = \sqrt{16 + 49} = \sqrt{65} \approx 8.06$

ANSWER

$\begin{pmatrix}4\\7\end{pmatrix}, \sqrt{65}$

EXAMPLE 2 **Show that $A(1, 1)$, $B(3, 4)$ and $C(7, 10)$ are collinear.**

1. $AB = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

2. $BC = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$

3. $BC = 2 \cdot AB$

   Parallel.

4. B lies on both, so the three are collinear.

   The concluding sentence.

ANSWER

Collinear, since $BC = 2 \cdot AB$ and B is common

EXAMPLE 3 **Find the unit vector in the direction of $\begin{pmatrix} -6 \\ 8 \end{pmatrix}$.**

1. $|v| = \sqrt{36 + 64} = 10$

2. $\frac{1}{10} \begin{pmatrix} -6 \\ 8 \end{pmatrix}$

3. $= \begin{pmatrix} -0.6 \\ 0.8 \end{pmatrix}$

   Check: $0.36 + 0.64 = 1$.

ANSWER

$\begin{pmatrix} -0.6 \\ 0.8 \end{pmatrix}$

EXAMPLE 4 **A triangle of area 12 is enlarged by factor -2 about $(1, 1)$. Describe the image.**

1. Lengths double.

2. Area × $k^2 = 4$, so 48.
   k^2 is positive either way.

3. The image is on the opposite side of $(1, 1)$, inverted.

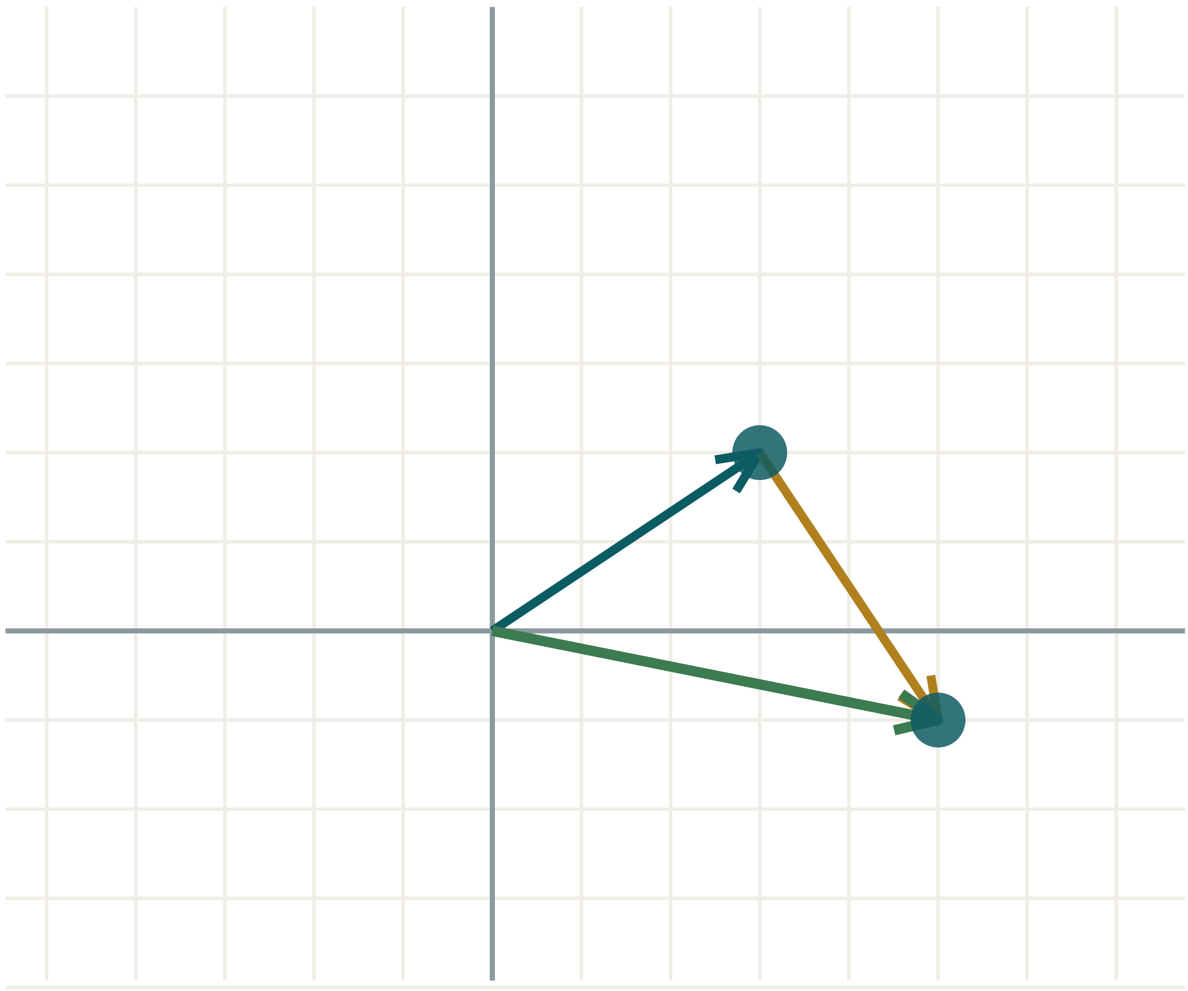
ANSWER

Area 48, inverted, on the far side of the centre

03 Interactive model

Cut a triangle from card and physically translate, reflect, rotate and enlarge it on squared paper.

Adding vectors nose to tail



a (3; 2)	b (2; -3)	a + b (5; -1)	|a| 3.61	|b| 3.61	|a + b| 5.1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Vector	Vektor	Вектор
Column vector	Ustun vektor	Вектор-столбец
Magnitude	Modul	Модуль
Position vector	Radius-vektor	Радиус-вектор
Resultant	Teng ta’sir etuvchi	Равнодействующая
Scalar multiple	Skalyarga ko‘paytma	Умножение на число
Translation	Parallel ko‘chirish	Параллельный перенос
Reflection	Simmetriya	Отражение
Rotation	Burish	Поворот
Enlargement	Gomotetiya	Гомотетия
Scale factor	O‘xshashlik koeffitsiyenti	Коэффициент подобия
Invariant point	O‘zgarmas nuqta	Неподвижная точка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

AB equals:

- $OA - OB$
- $OB - OA$
- $OA + OB$
- $|OB|$

$|(3,\ 4)|$ is:

- 7
- 5
- 12
- 25

Two vectors are parallel when:

- they are equal
- one is a scalar multiple of the other
- they have equal magnitude
- they meet

Collinearity needs, beyond parallel vectors:

- equal length
- a common point
- the origin
- nothing

A rotation is fully described by:

- the angle
- the centre and angle
- centre, angle and sense
- the centre

Under an enlargement of factor 3, areas multiply by:

- 3
- 6
- 9
- 27

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\begin{pmatrix}2\\3\end{pmatrix} + \begin{pmatrix}4\\1\end{pmatrix}$

ANSWER $\begin{pmatrix}6\\4\end{pmatrix}$

2. $\begin{pmatrix}5\\2\end{pmatrix} - \begin{pmatrix}1\\6\end{pmatrix}$

ANSWER $\begin{pmatrix}4\\-4\end{pmatrix}$

3. $3\begin{pmatrix}2\\-1\end{pmatrix}$

ANSWER $\begin{pmatrix}6\\-3\end{pmatrix}$

4. $|\begin{pmatrix}3\\4\end{pmatrix}|$

ANSWER 5

5. $|\begin{pmatrix}5\\12\end{pmatrix}|$

ANSWER 13

6. AB for $A(1, 2), B(5, 9)$

ANSWER $\begin{pmatrix}4\\7\end{pmatrix}$

7. Area factor of an enlargement with $k = 4$

ANSWER 16

Medium · the standard question

7 PROBLEMS

1. $|AB|$ for $A(1, 2), B(5, 9)$

ANSWER $\sqrt{65} \approx 8.06$

2. Unit vector along $\begin{pmatrix} -6 \\ 8 \end{pmatrix}$

ANSWER $\begin{pmatrix} -0.6 \\ 0.8 \end{pmatrix}$

3. Are $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 9 \\ 12 \end{pmatrix}$ parallel?

ANSWER Yes, $k = 3$

4. Are $A(1,1), B(3,4), C(7,10)$ collinear?

ANSWER Yes — $BC = 2 \cdot AB$

5. Midpoint of $A(2, 3)$ and $B(8, 11)$ as a position vector

ANSWER $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

6. Describe fully: $(1,1) \to (4,3)$ for every point

ANSWER Translation by $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

7. A shape of area 12 enlarged by $k = -2$: new area

ANSWER 48

Hard · stretch and reason

7 PROBLEMS

1. Find k so that $\begin{pmatrix} k \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

   ANSWER $k = 4$

2. $OP = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $OQ = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$: find the point dividing PQ in $1 : 2$

   ANSWER $(3, 3)$

3. Prove that the diagonals of a parallelogram bisect each other, using vectors

   ANSWER Both midpoints are $\frac{1}{2}(OA + OC))$

4. A rotation of $90°$ anticlockwise about O sends $(3, 1)$ to:

   ANSWER $(-1, 3)$

5. A reflection in $y = x$ sends $(4, -2)$ to:

   ANSWER $(-2, 4)$

6. An enlargement centre $(1, 1)$, factor 3 sends $(3, 2)$ to:

   ANSWER $(7, 4)$

7. Show that $ABCD$ with $A(0,0)$, $B(4,1)$, $C(6,5)$, $D(2,4)$ is a parallelogram

   ANSWER $AB = DC = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

07 Homework

Homework — 5 tasks

Every transformation answer must be a full description, not a name.

1. With $A(-1, 3)$ and $B(5, -5)$, find AB, $|AB|$ and the unit vector along AB.
2. Show that $P(2, 1)$, $Q(5, 7)$ and $R(9, 15)$ are collinear, and write the concluding sentence in full.
3. Find k so that $\begin{pmatrix} 6 \\ k \end{pmatrix}$ is parallel to $\begin{pmatrix} 4 \\ 10 \end{pmatrix}$.

4. Describe fully the transformation that sends $(1, 2)$ to $(-2, 1)$ and $(3, 2)$ to $(-2, 3)$.

5. A triangle of area 9 is enlarged by factor $\frac{2}{3}$ about the origin. Find the area of the image.

Home / Grade 10 / Geometry / Quarter 4

Control work 4, and work on the mistakes

Coordinate geometry in one paper, then the plane methods drawn as a single map.

LESSON 61–62 | 2 × 40 MIN | GEOMETRY 10, NAZORAT ISHI 4 | P1 3 REVIEW

LESSON CLOCK

3 | 37' | 11' | 20' | 9'

Instructions	3 min
The paper	37 min
Answers	11 min
Diagnosis and rewrite	20 min
The map	9 min

LEARNING OBJECTIVES

- Apply the line, circle and vector methods of Lessons 53–60 under time.
- Choose between the algebraic and the geometric route to a tangent.
- Classify each lost mark and rewrite the solution in full.
- Draw the coordinate block as one map.

TEXTBOOK

National	pp. 253–256
Cambridge	Pure Mathematics 1, pp. 73–76
Workbook	Control paper G4

01 Explanation

The paper — 35 marks, 42 minutes

Q	TASK	MARKS	FROM
1	$A(-1, 2)$, $B(7, 8)$: find $\lvert AB\rvert$, the midpoint and the gradient	*5*	L53–54
2	The same points: find the perpendicular bisector of AB	*5*	L53–54
3	Find the centre and radius of $x^2 + y^2 - 8x + 2y - 8 = 0$	*5*	L55–56
4	Find where $y = x - 1$ meets that circle, and the length of the chord	*7*	L57–58
5	Find the values of c for which $y = 2x + c$ is a tangent to $x^2 + y^2 = 45$	*6*	L57–58
6	Show that $P(1, 2)$, $Q(4, 8)$ and $R(7, 14)$ are collinear, using vectors	*7*	L59–60

WHERE THE MARKS ACTUALLY GO

Q2 carries one mark for the negative reciprocal and one for using the midpoint; Q3 one for the sign of the centre; Q4 one for giving both coordinates of each point; Q6 one for the concluding sentence. Five of the thirty-five marks are for a habit, not a calculation.

Naming the slip

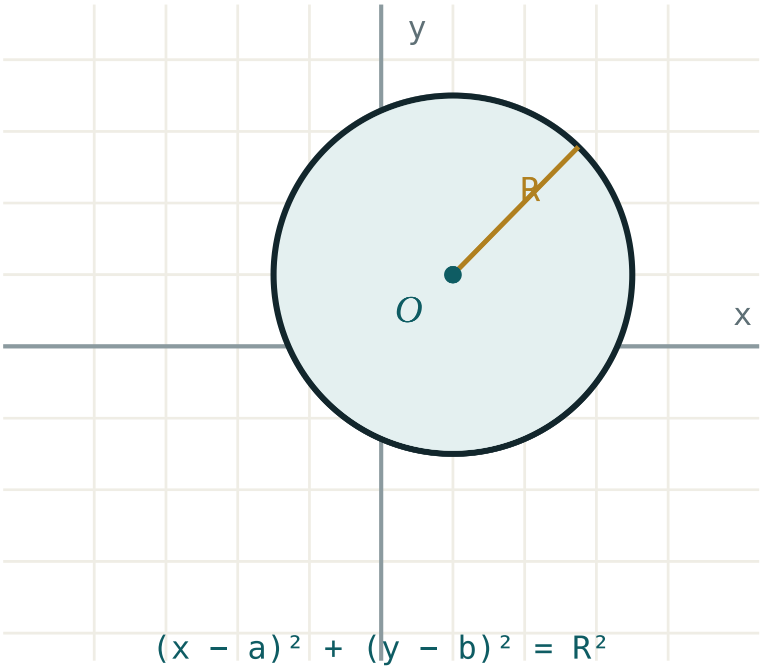
SLIP	WHAT IT LOOKS LIKE	THE FIX
gradient inverted	$\frac{x_2 - x_1}{y_2 - y_1}$	rise over run, in that order
centre sign	$(x - 3)^2 \Rightarrow \text{centre } -3$	reverse the bracket sign
half a point	$x = 3, x = -4$	substitute back for y
bisector not through M	right gradient, wrong line	two conditions, both needed
tangent by guessing	no D and no distance	either route, but write one
vector proof unfinished	$QR = 2 \cdot PQ$	add “and Q is common”
AB reversed	$OA - OB$	“to minus from”

Name the slip in the margin, then rewrite the whole question. A rewritten line is forgotten by Friday; a rewritten solution is not.

The coordinate block as one map

Six boxes, links written as sentences:

- **two points** → **length, midpoint, gradient** — “three formulas, one subtraction each”
- **gradient** → **parallel and perpendicular** — “equal, or negative reciprocal”
- **midpoint + perpendicular** → **perpendicular bisector** — “equidistant from both”
- **centre and radius** → **circle equation** — “the distance formula, squared”
- **substitute** → **quadratic** → **discriminant** — “two, one or no points”
- **vector** → **parallel + common point** — “collinear”



The centre-and-radius picture that half the paper rests on.

TWO ROUTES TO A TANGENT, AND WHEN TO USE EACH

The **discriminant** route always works but is longer. The **distance** route, $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} = r$, is one line — use it whenever the circle's centre is easy to write down.

Looking forward

Lessons 63–68 leave the plane and return to space: parallelism and perpendicularity, sections of polyhedra and orthogonal projection, then the annual paper. Nothing new is introduced — it is Quarters I to III, revisited with a year's practice behind them.

ONE HABIT TO CARRY FORWARD

Coordinate geometry rewarded drawing the points before calculating. Solid geometry rewards drawing the solid before naming the angle. In both, the sketch is where the marks are decided.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: the perpendicular bisector of $A(-1, 2)$ and $B(7, 8)$.**

1. $M = (3, 5)$

2. $m\_\{AB\} = \frac{6}{8} = \frac{3}{4}$

3. $m = -\frac{4}{3}$

   Negative reciprocal.

4. $y - 5 = -\frac{4}{3}(x - 3) \Rightarrow 4x + 3y = 27$

ANSWER

$4x + 3y = 27$

EXAMPLE 2 **Model answer, Q4: where $y = x - 1$ meets $x^2 + y^2 - 8x + 2y - 8 = 0$, and the chord length.**

1. Centre $(4, -1)$, $r = 5$.
   From Q3.

2. $x^2 + (x - 1)^2 - 8x + 2(x - 1) - 8 = 0 \Rightarrow 2x^2 - 10x - 9 = 0$

3. $x = \dfrac{10 \pm \sqrt{172}}{4}$, so $x \approx 5.78$ and $x \approx -0.78$.

4. Distance from the centre to the line $\dfrac{|4 - (-1) - 1|}{\sqrt{2}} = \dfrac{4}{\sqrt{2}}$; chord $2\sqrt{25 - 8} = 2\sqrt{17} \approx 8.25$.
   The faster route.

ANSWER

Chord $2\sqrt{17} \approx 8.25$

EXAMPLE 3 **Model answer, Q5: for which c is $y = 2x + c$ a tangent to $x^2 + y^2 = 45$?**

1. $r = \sqrt{45} = 3\sqrt{5}$

2. Distance from $(0, 0)$ to $2x - y + c = 0$ is $\dfrac{|c|}{\sqrt{5}}$.

3. $\dfrac{|c|}{\sqrt{5}} = 3\sqrt{5} \Rightarrow |c| = 15$

4. $c = \pm 15$

ANSWER

$c = 15$ or $c = -15$

03 Interactive model

Work Q5 both ways on the board — discriminant and distance — and time each; the class will not use the long one again.

Ten questions on the coordinate block

Gradient through $(1, 2)$ and $(5, 10)$:

- 2
- $\frac{1}{2}$
- 8
- 4

Perpendicular to gradient $\frac{3}{4}$:

- $\frac{4}{3}$
- $-\frac{4}{3}$
- $-\frac{3}{4}$
- $\frac{3}{4}$

Centre of $(x + 2)^2 + (y - 5)^2 = 9$:

- $(2, -5)$
- $(-2, 5)$
- $(2, 5)$
- $(-2, -5)$

Its radius:

- 9
- 3
- 81
- $\sqrt{3}$

A line meets a circle in at most:

- 1
- 2
- 3
- 4

$D = 0$ means:

- a secant
- a tangent
- no meeting
- a diameter

The chord at distance d:

- $\sqrt{r^2 - d^2}$
- $2\sqrt{r^2 - d^2}$
- $2d$
- $r + d$

AB equals:

- $OA - OB$
- $OB - OA$
- $OA + OB$
- $|AB|$

Collinear needs, beyond parallel:

- equal length
- a common point
- the origin
- nothing

An enlargement of factor k multiplies areas by:

- k
- k^2
- k^3
- $2k$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Coordinate method	Koordinatalar usuli	Координатный метод
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Discriminant	Diskriminant	Дискриминант
Tangent	Urinma	Касательная
Position vector	Radius-vektor	Радиус-вектор
Concept map	Tushunchalar xaritasi	Карта понятий
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q2 needs two things:

- two points
- the midpoint and the perpendicular gradient
- the length
- the radius

The faster route in Q5 is:

- the discriminant
- the distance from the centre
- guessing
- a sketch

A vector proof of collinearity ends with:

- the ratio
- a sentence naming the common point
- the magnitude
- nothing

Lessons 63–68 return to:

- circles
- solid geometry
- probability
- vectors in the plane

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $|AB|$ for $A(-1, 2), B(7, 8)$

   ANSWER 10

2. Midpoint of the same

   ANSWER $(3, 5)$

3. Gradient of the same

   ANSWER $\frac{3}{4}$

4. Centre of $x^2 + y^2 - 8x + 2y - 8 = 0$

   ANSWER $(4, -1)$

5. Radius of the same

   ANSWER 5

6. Gradient perpendicular to $\frac{3}{4}$

   ANSWER $-\frac{4}{3}$

7. PQ for $P(1, 2), Q(4, 8)$

   ANSWER $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$

Medium · the standard question

7 PROBLEMS

1. Perpendicular bisector of $A(-1, 2)$ and $B(7, 8)$

ANSWER $4x + 3y = 27$

2. Chord of $x^2 + y^2 - 8x + 2y - 8 = 0$ on $y = x - 1$

ANSWER $2\sqrt{17} \approx 8.25$

3. Tangency values of c for $y = 2x + c$ and $x^2 + y^2 = 45$

ANSWER $c = \pm 15$

4. Are $P(1,2)$, $Q(4,8)$, $R(7,14)$ collinear?

ANSWER Yes — $QR = PQ$, Q common

5. Line through $(2, 3)$ parallel to $4x + 3y = 1$

ANSWER $4x + 3y = 17$

6. Distance from $(4, -1)$ to $y = x - 1$

ANSWER $\frac{4}{\sqrt{2}} = 2\sqrt{2}$

7. Circle centre $(4, -1)$ radius 5, in expanded form

ANSWER $x^2 + y^2 - 8x + 2y - 8 = 0$

Hard · stretch and reason

7 PROBLEMS

1. Tangent to $x^2 + y^2 = 45$ at $(6, 3)$

   ANSWER $6x + 3y = 45$, i.e. $2x + y = 15$

2. Circumcentre of $(-1, 2), (7, 8), (7, 2)$

   ANSWER $(3, 5)$

3. The line $4x + 3y = 27$ and the circle $x^2 + y^2 = 25$: how many points?

   ANSWER None — $d = 5.4 > 5$

4. Area of the triangle $(0, 0), (8, 0), (3, 6)$

   ANSWER 24

5. Find k so that $\begin{pmatrix} k \\ 9 \end{pmatrix}$ is parallel to $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$

   ANSWER $k = 3$

6. Shortest distance from $(10, 5)$ to the circle $(x - 4)^2 + (y + 1)^2 = 25$

   ANSWER $\sqrt{72} - 5 \approx 3.49$

7. Show that the perpendicular bisectors of the sides of $(0,0), (6,0), (0,8)$ meet at $(3, 4)$

   ANSWER All three pass through it

07 Homework

Homework — 4 tasks

Bring the concept map to Lesson 63; the solid-geometry revision assumes the plane work is secure.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Complete the concept map with all six links written as sentences.
3. Find the values of c for which $y = x + c$ is a tangent to $x^2 + y^2 = 18$, by both routes.
4. Write your target for the revision block in one checkable sentence, and date it.

Revision — parallelism and perpendicularity in space

The whole year of solid geometry, reduced to eight statements and one picture each.

LESSON 63–64 | 2 × 40 MIN | GEOMETRY 10, CHAPTERS II–IV REVIEW | P1 SPATIAL REVIEW

LESSON CLOCK

12' | 20' | 20' | 16' | 12'

Positions	12 min
Parallelism	20 min
Perpendicularity	20 min
Angles	16 min
Homework	12 min

LEARNING OBJECTIVES

- State the positions of two lines, a line and a plane, and two planes.
- Use the criteria for a line parallel to a plane and for two parallel planes.
- Use the criterion for a line perpendicular to a plane, and the three-perpendiculars theorem.
- Compute the angle between a line and a plane and between two planes.

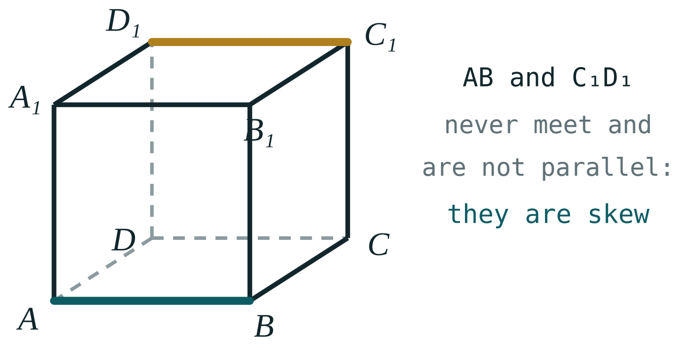
TEXTBOOK

National	pp. 257–266
Cambridge	Pure Mathematics 1, pp. 77–80
Workbook	Revision set G1

01 Explanation

Positions

PAIR	POSSIBLE POSITIONS
two lines	intersecting · parallel · **skew**
a line and a plane	the line lies in it · parallel · meets it in one point
two planes	coincident · parallel · meeting in a line



Skew lines — not parallel, and never meeting. The case that has no analogue in the plane.

SKEW IS THE CASE THAT ONLY SPACE HAS

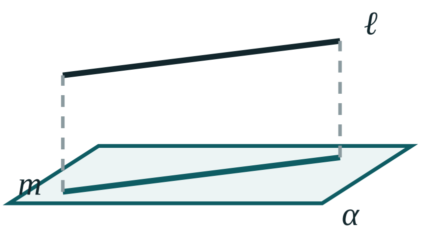
Two lines that do not meet need not be parallel — they may simply lie in different planes. In a cube the twelve edges give 18 parallel pairs, 24 intersecting pairs and 24 skew pairs, and $18 + 24 + 24 = 66 = C(12, 2)$.

Parallelism

THE TWO CRITERIA

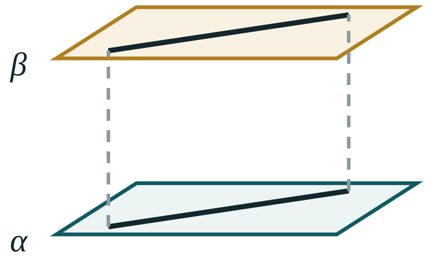
- A line is **parallel to a plane** if it is parallel to some line lying in that plane.
- Two planes are **parallel** if two intersecting lines of one are parallel to two intersecting lines of the other.

In both, the word **intersecting** is essential: two parallel lines of one plane parallel to two of the other prove nothing.



$\ell \parallel m$ and m lies in α, so $\ell \parallel \alpha$

A line parallel to a plane — it is parallel to a line drawn in the plane.



$\alpha \parallel \beta$ — a third plane cuts them in parallel lines

Two intersecting lines of each, in parallel pairs — that is the criterion.

Two consequences used constantly:

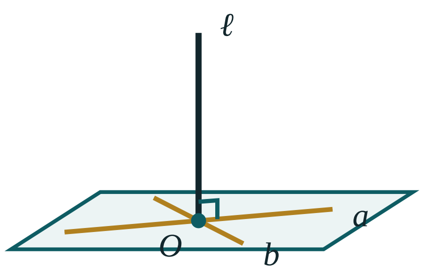
- If $\alpha \parallel \beta$ and a third plane cuts both, the two lines of intersection are parallel.
- Segments of parallel lines between parallel planes are equal.

Perpendicularity

THE CRITERION, AND THE THEOREM

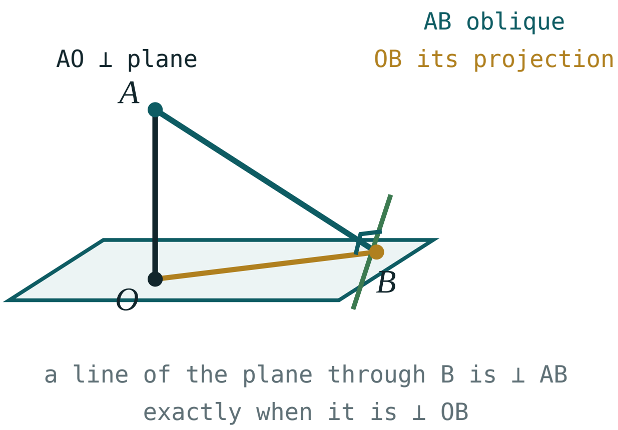
A line is **perpendicular to a plane** if it is perpendicular to two **intersecting** lines of that plane. It is then perpendicular to **every** line of the plane.

> **Three perpendiculars.** A line in a plane is perpendicular to an oblique if and only if it is perpendicular to the oblique’s projection.



$\ell \perp a$ and $\ell \perp b$, with a and b intersecting at O
therefore ℓ is perpendicular to the whole plane

Perpendicular to two intersecting lines — and therefore to all of them.

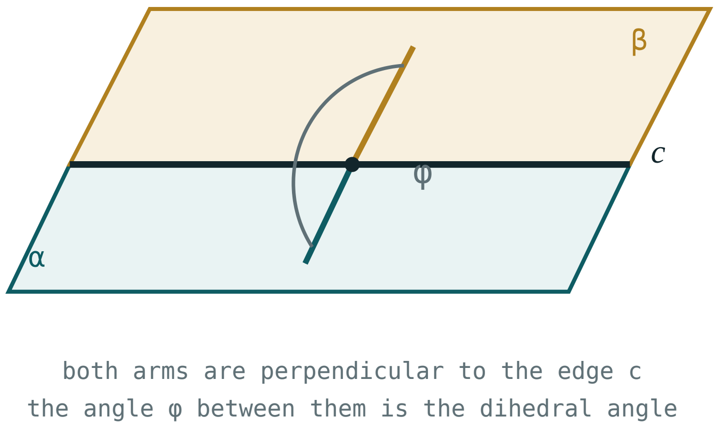


The oblique, its projection, and the line in the plane — perpendicular to one iff to the other.

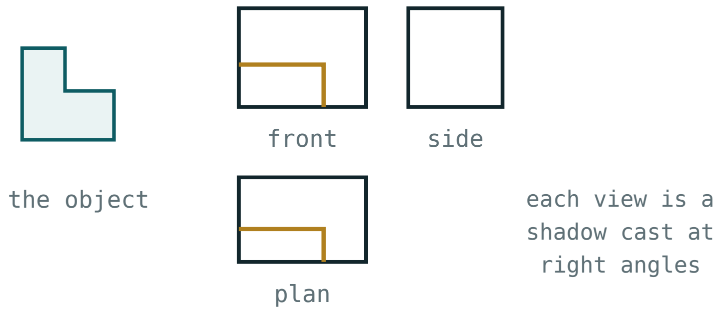
The three-perpendiculars theorem is what turns a space problem into a plane one: it lets you drop from a slanted segment to its shadow and work entirely in the base.

Angles

ANGLE	HOW TO CONSTRUCT IT
line and plane	between the line and its **orthogonal projection**
two planes (dihedral)	between two rays perpendicular to the edge, one in each face
two skew lines	translate one until they meet; the angle between the translates



The linear angle of a dihedral — both rays perpendicular to the edge.



three orthogonal projections determine the solid

The projection is the shadow cast straight down; the angle is measured to it.

EVERY ANGLE QUESTION BECOMES ONE RIGHT TRIANGLE

Find the foot of the perpendicular, name the projection, and a right triangle appears with the slant as hypotenuse. Then $\cos\theta = \dfrac{\text{projection}}{\text{slant}}$ or $\tan\theta = \dfrac{\text{height}}{\text{projection}}$ — plane trigonometry from that point on.

02 Worked examples

EXAMPLE 1 **In a cube $ABCDA_1B_1C_1D_1$ of edge a, find the angle between AC_1 and the base.**

1. The projection of AC_1 on the base is AC.
2. $AC = a\sqrt{2}$, $CC_1 = a$
3. $\tan\theta = \dfrac{a}{a\sqrt{2}} = \dfrac{1}{\sqrt{2}}$
4. $\theta \approx 35.3°$

ANSWER

$\approx 35.3°$

EXAMPLE 2 **In the same cube, find the angle between the planes ABC_1D_1 and the base.**

1. The edge of the dihedral is AB.
2. $BC \perp AB$ in the base; $BC_1 \perp AB$ in the upper face.
   Both perpendicular to the edge.
3. $\tan\theta = \dfrac{CC_1}{BC} = 1$
4. $\theta = 45°$

ANSWER

$45°$

EXAMPLE 3 **Prove that a line perpendicular to two intersecting lines of a plane is perpendicular to a third line of that plane.**

1. Let $l \perp a$ and $l \perp b$, with $a \cap b = O$.
2. Any line c of the plane through O is a combination of directions of a and b.
3. So $l \perp c$ as well.
   This is the criterion, and its point.

ANSWER

Perpendicular to two intersecting lines means perpendicular to all

EXAMPLE 4

A point is *12* cm from a plane. Two obliques of lengths *13* and *15* cm are drawn. Find their projections.

1. $\sqrt{169 - 144} = 5$

2. $\sqrt{225 - 144} = 9$

3. Projections 5 cm and 9 cm.

   The longer oblique has the longer projection.

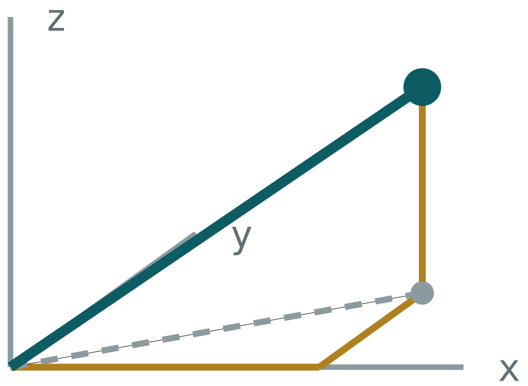
ANSWER

5 cm and 9 cm

03 Interactive model

Use a wire cube and a torch: the shadow on the desk is the orthogonal projection, and every angle question becomes visible.

Lines and planes in a box



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Skew lines	Ayqash to‘g‘ri chiziqlar	Скрещивающиеся прямые
Parallel to a plane	Tekislikka parallel	Параллельная плоскости
Parallel planes	Parallel tekisliklar	Параллельные плоскости
Perpendicular to a plane	Tekislikka perpendikulyar	Перпендикулярная плоскости
Three perpendiculars theorem	Uch perpendikulyar teoremasi	Теорема о трёх перпендикулярах
Angle between a line and a plane	To‘g‘ri chiziq va tekislik orasidagi burchak	Угол между прямой и плоскостью
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Orthogonal projection	Ortogonal proyeksiya	Ортогональная проекция
Oblique	Og‘ma	Наклонная
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Two lines that never meet are:

- always parallel
- parallel or skew
- always skew
- intersecting

A line is perpendicular to a plane if it is perpendicular to:

- one line of it
- two intersecting lines of it
- two parallel lines of it
- the edge

The angle between a line and a plane is measured to:

- any line of the plane
- its orthogonal projection
- the normal
- the edge

A dihedral angle is measured by:

- any two rays
- two rays perpendicular to the edge
- the edge itself
- the projection

The three-perpendiculars theorem connects:

- two planes
- an oblique and its projection
- two skew lines
- two obliques

In a cube, the number of skew edge pairs is:

- 18
- 24
- 12
- 66

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up   7 PROBLEMS

1. Three positions of two lines in space

   ANSWER Intersecting, parallel, skew

2. Three positions of a line and a plane

   ANSWER In it, parallel, meeting in a point

3. A line ⊥ to two intersecting lines of a plane is:

   ANSWER Perpendicular to the plane

4. The angle between a line and a plane is measured to:

   ANSWER Its projection

5. Space diagonal of a cube of edge a

   ANSWER $a\sqrt{3}$

6. Face diagonal of a cube of edge a

   ANSWER $a\sqrt{2}$

7. Projection of an oblique 13 from a point 12 from the plane

   ANSWER 5

Medium · the standard question

7 PROBLEMS

1. Angle between AC_1 and the base in a cube

ANSWER $\approx 35.3°$

2. Angle between ABC_1D_1 and the base in a cube

ANSWER $45°$

3. Angle between two space diagonals of a cube

ANSWER $\approx 70.5°$ or $\approx 109.5°$

4. Number of parallel edge pairs in a cube

ANSWER 18

5. Number of intersecting edge pairs in a cube

ANSWER 24

6. Number of skew edge pairs in a cube

ANSWER 24

7. Two obliques of 15 and 20 from a point 12 from a plane: projections

ANSWER 9 and 16

Hard · stretch and reason

7 PROBLEMS

1. Distance from a vertex of a cube of edge a to a space diagonal not through it

ANSWER $\frac{a\sqrt{6}}{3}$

2. Angle between AB_1 and BC_1 in a cube

ANSWER $60°$

3. Distance between two skew edges of a cube of edge a

ANSWER a

4. A rectangular box $3 \times 4 \times 12$: the angle of the space diagonal with the base

ANSWER $\approx 67.4°$

5. Prove that if $\alpha \parallel \beta$ and γ cuts both, the lines of intersection are parallel

ANSWER They lie in γ and cannot meet

6. A point 6 from a plane: the locus of feet of obliques of length 10

ANSWER A circle of radius 8

7. A regular tetrahedron of edge a: the angle between a lateral edge and the base

ANSWER $\approx 54.7°$

07 Homework

Homework — 5 tasks

Draw the cube for every question; the marks live in the picture.

1. List the three positions of two lines, of a line and a plane, and of two planes, with one sketch each.
2. In a cube of edge 6 cm, find the angle between the space diagonal and the base.
3. In the same cube, find the dihedral angle between the plane through A, B, C_1 and the base.
4. A point is 8 cm from a plane; an oblique from it is 17 cm long. Find the projection.

5. State the three-perpendiculars theorem and draw the picture that goes with it.

Revision — sections of polyhedra and orthogonal projection

Three rules build any section; one shadow measures any angle.

LESSON 65–66 | 2 × 40 MIN | GEOMETRY 10, CHAPTER IV REVIEW | P1 SPATIAL REVIEW

LESSON CLOCK

12' | 24' | 16' | 16' | 12'

Three rules	12 min
Building a section	24 min
Naming the shape	16 min
Areas by projection	16 min
Homework	12 min

LEARNING OBJECTIVES

- Construct the section of a cube or prism by a plane through three points.
- Use the trace of a cutting plane on a face of the solid.
- Compute the area of a section by projecting it onto the base.
- Use orthogonal projection to find lengths and angles.

TEXTBOOK

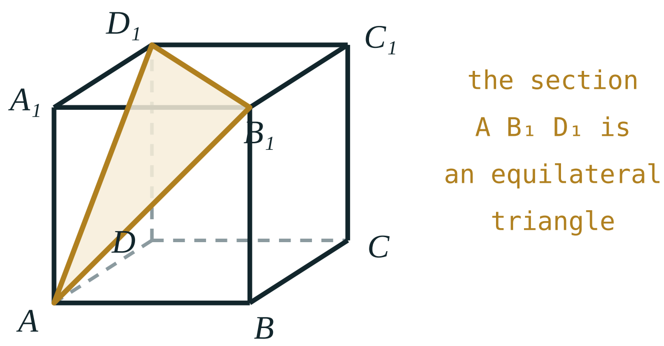
National	pp. 267–276
Cambridge	Pure Mathematics 1, pp. 81–84
Workbook	Revision set G2

01 Explanation

Three rules

EVERYTHING A SECTION NEEDS

1. Two points of the cutting plane on the **same face** join up — the segment between them is part of the section.
2. The cutting plane meets two **parallel faces** in **parallel** lines.
3. Where the cutting plane meets the plane of a face outside the solid, that point is on the **trace**, and can be used to reach a further face.



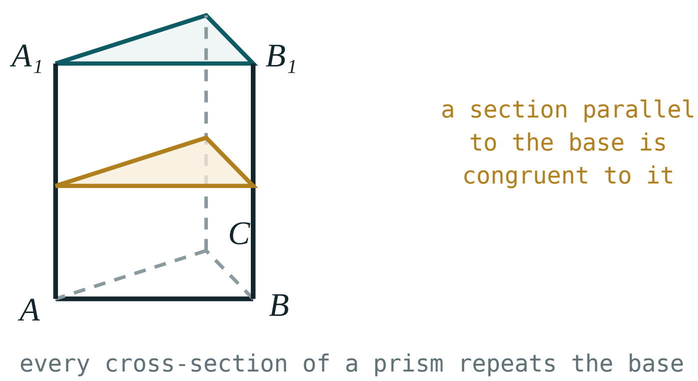
A plane through three points of a cube — each segment lies in one face.

> A SECTION IS A POLYGON, AND ITS SIDES LIE IN FACES
>
> Every side of a section is the intersection of the cutting plane with **one face**. A drawn line that crosses from one face to another is not a side of anything — it is a mistake.

Building a section

The method, for a plane through three given points of a cube:

1. join any two points that lie on the same face — rule 1;
2. extend that line to meet the plane of an adjacent face outside the solid — rule 3;
3. join the new point to the third given point, and take the part inside the solid;
4. use rule 2 on parallel faces to finish, and check every side lies in exactly one face.



The same three rules on a prism — a plane cut across the lateral faces.

CUT OF A CUBE	SHAPE	NOTE
parallel to a face	square	congruent to the face
through two opposite edges	rectangle	a diagonal section
through three face-diagonals	equilateral triangle	side $a\sqrt{2}$
through six edge-midpoints	regular hexagon	side $\frac{a\sqrt{2}}{2}$

Naming the shape

The shape of the section is decided by how many faces the plane crosses — one side per face crossed.

FACES CROSSED	SECTION	EXAMPLE IN A CUBE
3	triangle	a corner cut off
4	quadrilateral	a plane parallel to an edge
5	pentagon	an oblique cut
6	hexagon	through six edge-midpoints

A cube has six faces, so its sections have at most six sides. A tetrahedron has four, so its sections are triangles or quadrilaterals — never more.

PARALLEL FACES FORCE PARALLEL SIDES

In a cube the faces come in three parallel pairs, so a hexagonal section has its opposite sides parallel — and the regular one, through the six edge-midpoints, has all six sides equal as well.

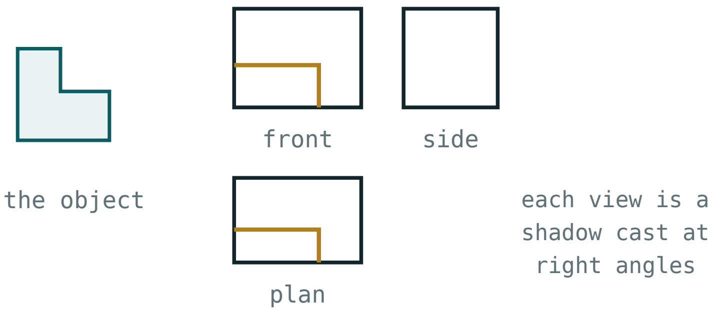
Areas by projection

When a section is awkward to measure directly, project it onto the base.

$$S\_projection = S\_section \cdot \cos\theta$$

where θ is the dihedral angle between the two planes. Read backwards, this finds the section from a shape that is easy to measure:

$$\mathit{S\_section} = \frac{\mathrm{S\_projection}}{\cos\theta}$$



three orthogonal projections determine the solid

The shadow is shorter than the object by exactly the factor cos θ.

Example. A plane cuts a cube of edge 6 through A, B and C_1. Its projection on the base is the triangle ABC, of area 18; the dihedral angle is $45°$; so the section has area $\frac{18}{\cos 45°} = 18\sqrt{2} \approx 25.5$.

THE PROJECTION IS ALWAYS THE SMALLER

Since $\cos\theta \le 1$, a shadow is never larger than what casts it. If your section comes out *smaller* than its projection, the formula has been used upside down.

02 Worked examples

EXAMPLE 1 **A plane passes through the midpoints of six edges of a cube of edge *a*. Name the section and find its perimeter.**

1. Six faces crossed, so a hexagon.
2. Each side joins two midpoints of one face: $\frac{a\sqrt{2}}{2}$.
3. All six equal — a regular hexagon.
4. Perimeter $3a\sqrt{2}$

ANSWER

A regular hexagon of perimeter $3a\sqrt{2}$

EXAMPLE 2 **Find the area of the diagonal section of a cube of edge *6*.**

1. It is the rectangle through two opposite edges.
2. Sides 6 and $6\sqrt{2}$.
3. $S = 36\sqrt{2} \approx 50.9$

ANSWER

$36\sqrt{2} \approx 50.9$

EXAMPLE 3 **A section of area S makes $60°$ with the base. Find its projection.**

1. $S\_proj = S \cos 60°$
2. $= 0.5\ S$
3. Half the section.

ANSWER

$0.5\ S$

EXAMPLE 4 **A plane through A, B and C_1 cuts a cube of edge 6. Find the area of the section.**

1. Projection on the base: triangle ABC, area 18.
2. Dihedral angle along AB is $45°$.
3. $S = \dfrac{18}{\cos 45°}$
4. $= 18\sqrt{2} \approx 25.5$

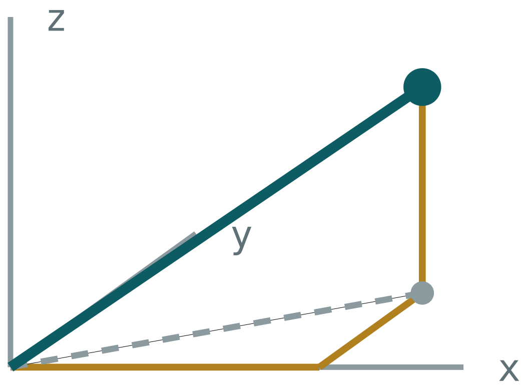
ANSWER

$18\sqrt{2} \approx 25.5$

03 Interactive model

Cut a potato or a block of foam along a plane through three marked points; the class predicts the polygon before the cut.

Cutting the box



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Section (cross-section)	Kesim	Сечение
Cutting plane	Kesuvchi tekislik	Секущая плоскость
Trace	Iz	След
Polyhedron	Ko‘pyoq	Многогранник
Face	Yoq	Грань
Edge	Qirra	Ребро
Orthogonal projection	Ortogonal proyeksiya	Ортогональная проекция
Projection of an area	Yuza proyeksiyasi	Проекция площади
Diagonal section	Diagonal kesim	Диагональное сечение
Midpoint section	O‘rta kesim	Среднее сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Each side of a section lies in:

- two faces
- exactly one face
- the base
- no face

A cutting plane meets two parallel faces in:

- perpendicular lines
- parallel lines
- one point
- nothing

A cube’s section has at most:

- 4 sides
- 5 sides
- 6 sides
- 8 sides

A tetrahedron's section has at most:

- 3
- 4
- 5
- 6

$S\_projection$ equals:

- $S \cos \theta$
- $\frac{S}{\cos \theta}$
- $S \sin \theta$
- S

A section through six edge-midpoints of a cube is:

- a square
- a regular hexagon
- a triangle
- a pentagon

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. A plane parallel to a face of a cube cuts:

   ANSWER A square

2. A corner cut off a cube gives:

   ANSWER A triangle

3. Maximum number of sides of a cube section

   ANSWER 6

4. Maximum number of sides of a tetrahedron section

   ANSWER 4

5. Side of the equilateral section through three face-diagonals of a cube of edge a

   ANSWER $a\sqrt{2}$

6. Diagonal section of a cube of edge 6

   ANSWER $36\sqrt{2}$

7. $S\_projection$ when $S = 20, \theta = 60°$

   ANSWER 10

Medium · the standard question 7 PROBLEMS

1. Perimeter of the regular hexagonal section of a cube of edge a

ANSWER $3a\sqrt{2}$

2. Area of that hexagon for $a = 4$

ANSWER $12\sqrt{3} \approx 20.8$

3. Area of the equilateral section through three face-diagonals, $a = 6$

ANSWER $18\sqrt{3} \approx 31.2$

4. Section through A, B, C_1 of a cube of edge 6

ANSWER $18\sqrt{2} \approx 25.5$

5. $S\_section$ when $S\_proj = 12, \theta = 45°$

ANSWER $12\sqrt{2} \approx 17.0$

6. How many faces does a pentagonal section cross?

ANSWER 5

7. A plane parallel to two opposite edges of a cube cuts:

ANSWER A rectangle

Hard · stretch and reason

7 PROBLEMS

1. Prove that the hexagonal section through the six edge-midpoints of a cube is regular

   ANSWER Each side is $\frac{a\sqrt{2}}{2}$, angles $120°$

2. Area of the section of a cube of edge a through one edge and the midpoint of the opposite edge

   ANSWER $\frac{a^2\sqrt{5}}{2}$

3. A plane cuts a cube of edge 4 through three vertices of one face and one of the opposite: name the section

   ANSWER A quadrilateral (a trapezium)

4. A section of a cube has area $12\sqrt{2}$ and projects to 12: find θ

   ANSWER $45°$

5. A regular tetrahedron of edge a: the area of the section through one edge and the midpoint of the opposite

   ANSWER $\frac{a^2\sqrt{2}}{4}$

6. A plane parallel to the base cuts a pyramid halfway up: the ratio of the areas

   ANSWER $1 : 4$

7. Show that the section of a cube through three face-diagonals meeting at a vertex is equilateral

   ANSWER All three sides are $a\sqrt{2}$

07 Homework

Homework — 5 tasks

Draw each cube large, and mark the three given points before drawing any line.

1. State the three rules for constructing a section, with a sketch for each.

2. Construct the section of a cube by the plane through the midpoints of six edges, and name it.
3. Find the area of the diagonal section of a cube of edge 8 cm.
4. A section makes $30°$ with the base and its projection has area 9 cm². Find the area of the section.
5. Explain in two sentences why the section of a tetrahedron can never be a pentagon.

Annual control work, and the Grade 11 preview

The whole year in one paper, then a look at the road that leads out of it.

LESSON 67–68 | 2 × 40 MIN | GEOMETRY 10, YILLIK NAZORAT ISHI | ANNUAL REVIEW

LESSON CLOCK

3 | 45' | 12' | 12' | 8'

Instructions	3 min
The paper	45 min
Answers	12 min
Diagnosis	12 min
The Grade 11 preview	8 min

LEARNING OBJECTIVES

- Apply the axiomatic, parallel, perpendicular and coordinate methods of the whole year.
- Choose the shortest route to each answer under time pressure.
- Classify each lost mark and rewrite the solution.
- See how Grade 11 continues each strand of Grade 10.

TEXTBOOK

National	pp. 277–280
Cambridge	Pure Mathematics 1, pp. 85–88
Workbook	Annual paper G

01 Explanation

The paper — 40 marks, 45 minutes

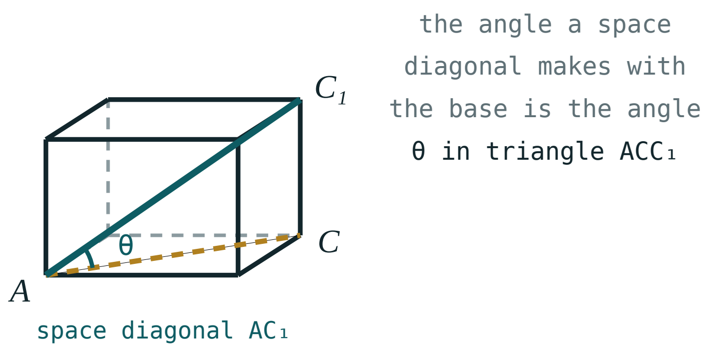
Q	TASK	MARKS	FROM
1	State the three positions of two lines in space, with a sketch of each	*4*	Q I
2	Prove that a line parallel to a line of a plane is parallel to the plane, or lies in it	*6*	Q II
3	A cube of edge 8: the angle between the space diagonal and the base, and between the plane ABC_1 and the base	*7*	Q III
4	Construct and name the section of a cube through three given vertices; find its area	*7*	Q III–IV
5	$A(2, -1)$, $B(8, 7)$: the length, the midpoint and the perpendicular bisector	*6*	Q IV
6	Find where $y = 2x - 4$ meets $x^2 + y^2 = 16$, and the chord length	*6*	Q IV
7	A point 9 cm from a plane; obliques of 15 and 41 cm: the projections	*4*	Q III

HOW TO SPEND 45 MINUTES

Questions 1, 5 and 7 are quick — about *12* minutes for *14* marks. Do them first, then 3 and 6, and leave 2 and 4 last: a proof and a construction reward the time that is left, and punish the time that is not.

The year in one page

QUARTER	THE ONE IDEA	THE PICTURE
I — axioms and entry to space	three points determine a plane	the plane through three dots
II — parallelism	parallel to a line **in** the plane	a line and its parallel below
III — perpendicularity	perpendicular to two **intersecting** lines	the vertical mast
IV — coordinates	geometry becomes algebra	the circle and its chord



One box, and every angle of the year measured inside it.

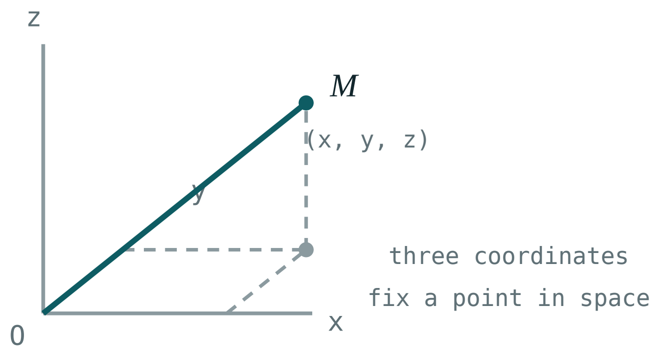
THE FOUR HABITS WORTH KEEPING

- draw the solid before naming anything;
- find the projection, and a right triangle appears;
- in coordinates, plot before you calculate;
- finish a proof with the sentence that states the conclusion.

The Grade 11 preview

Grade 11 geometry continues each of the four strands, and adds measurement.

GRADE 10	BECOMES, IN GRADE 11
points and lines in space	coordinates (x, y, z) and vectors in space
angles by projection	the scalar product, and angles without a picture
polyhedra and their sections	prisms and pyramids — surface area and volume
the circle in the plane	the cylinder, the cone and the sphere



The next step — a third coordinate, and every plane method carries over.

What to do over the summer. Nothing large. Keep the seven plane-geometry formulas and the four positions-in-space pictures somewhere you will see them, and in September the first lesson — the coordinate system in space — will read as an easy extension rather than a new subject.

ONE SENTENCE TO CARRY INTO GRADE 11

Every three-dimensional problem is solved by finding the one right triangle it contains. That was true in every lesson this year, and it stays true in every lesson of the next.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: a cube of edge *8* — the angle between the space diagonal and the base.**

1. Projection of AC_1 is $AC = 8\sqrt{2}$.
2. $CC_1 = 8$
3. $\tan\theta = \dfrac{8}{8\sqrt{2}} = \dfrac{1}{\sqrt{2}}$
4. $\theta \approx 35.3°$

ANSWER

$\approx 35.3°$

EXAMPLE 2 **Model answer, Q6: where $y = 2x - 4$ meets $x^2 + y^2 = 16$.**

1. $x^2 + (2x - 4)^2 = 16$
2. $5x^2 - 16x = 0 \Rightarrow x(5x - 16) = 0$
3. $x = 0$ or $x = 3.2$
4. $(0, -4)$ and $(3.2, 2.4)$; chord $\sqrt{10.24 + 40.96} \approx 7.16$.

ANSWER

$(0, -4)$, $(3.2, 2.4)$; chord ≈ 7.16

EXAMPLE 3 **Model answer, Q7: a point *9* cm from a plane, obliques *15* and *41* cm.**

1. $\sqrt{225 - 81} = 12$
2. $\sqrt{1681 - 81} = 40$
3. Projections 12 cm and 40 cm.

   Both are Pythagorean triples.

ANSWER

12 cm and 40 cm

03 Interactive model

Give the class the paper’s mark allocation before they start; ten seconds of planning is worth five marks.

The year in twelve questions

A plane is determined by:

- two points
- three non-collinear points
- one line
- four points

Two lines that never meet are:

- parallel
- parallel or skew
- skew
- intersecting

A line ‖ to a plane is ‖ to:

- every line of it
- some line of it
- the normal
- nothing

Two planes are parallel if:

- one line of each is parallel
- two intersecting lines of each are parallel
- they do not meet at one point
- they look parallel

⊥ to a plane means ⊥ to:

- one line
- two intersecting lines
- two parallel lines
- the edge

The angle to a plane is measured to:

- the normal
- the projection
- any line
- the edge

A dihedral angle is measured:

- anywhere
- by rays ⊥ to the edge
- along the edge
- by projection

Each side of a section lies in:

- one face
- two faces
- the base
- none

S_proj equals:

- $S \cos \theta$
- $\dfrac{S}{\cos \theta}$
- $S \sin \theta$
- S

Perpendicular gradients satisfy:

- $m_1 = m_2$
- $m_1 m_2 = -1$
- $m_1 + m_2 = 0$
- $m_1 m_2 = 1$

Centre of $(x - 4)^2 + (y + 1)^2 = 25$:

- $(4, 1)$
- $(4, -1)$
- $(-4, 1)$
- $(-4, -1)$

A tangent line meets the radius at:

- $45°$
- $90°$
- $60°$
- any angle

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Annual control work	Yillik nazorat ishi	Годовая контрольная работа
Axiom	Aksioma	Аксиома
Parallelism	Parallellik	Параллельность
Perpendicularity	Perpendikulyarlik	Перпендикулярность
Section	Kesim	Сечение
Coordinate method	Koordinatalar usuli	Координатный метод
Solid of revolution	Aylanish jismi	Тело вращения
Volume	Hajm	Объём

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which questions should be attempted first?

- the proof
- the quick ones — 1, 5 and 7
- the construction
- in order

Grade 11 begins geometry with:

- the sphere
- coordinates and vectors in space
- sections
- the circle

The one habit worth keeping is:

- memorising formulas
- finding the right triangle
- working fast
- skipping proofs

The scalar product will replace:

- the section method
- angles found by projection
- the axioms
- coordinates

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Three positions of two lines in space

   ANSWER Intersecting, parallel, skew

2. A plane is determined by

   ANSWER Three non-collinear points

3. Space diagonal of a cube of edge 8

   ANSWER $8\sqrt{3}$

4. Face diagonal of the same cube

   ANSWER $8\sqrt{2}$

5. Projection of a 15 oblique from a point 9 from the plane

   ANSWER 12

6. Midpoint of $(2, -1)$ and $(8, 7)$

   ANSWER $(5, 3)$

7. Length of that segment

   ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Angle of the space diagonal of a cube with the base

   ANSWER $\approx 35.3°$

2. Dihedral angle of the plane ABC_1 with the base of a cube

   ANSWER $45°$

3. Perpendicular bisector of $(2, -1)$ and $(8, 7)$

   ANSWER $3x + 4y = 27$

4. Where $y = 2x - 4$ meets $x^2 + y^2 = 16$

   ANSWER $(0, -4), (3.2, 2.4)$

5. Chord length there

   ANSWER ≈ 7.16

6. Projection of a 41 oblique from a point 9 from a plane

   ANSWER 40

7. Diagonal section of a cube of edge 8

   ANSWER $64\sqrt{2}$

Hard · stretch and reason

7 PROBLEMS

1. Area of the section of a cube of edge 8 through A, B, C_1

   ANSWER $32\sqrt{2} \approx 45.3$

2. Angle between two space diagonals of a cube

   ANSWER $\approx 70.5°$

3. Distance from a vertex of a cube of edge 8 to the opposite space diagonal

   ANSWER $\dfrac{8\sqrt{6}}{3} \approx 6.53$

4. A tangent to $x^2 + y^2 = 16$ parallel to $y = 2x$

   ANSWER $y = 2x \pm 4\sqrt{5}$

5. The regular hexagonal section of a cube of edge 8: its area

   ANSWER $48\sqrt{3} \approx 83.1$

6. A rectangular box $9 \times 12 \times 20$: the angle of the space diagonal with the base

   ANSWER $\approx 53.1°$

7. Prove that the four space diagonals of a cube meet at one point and bisect each other

   ANSWER All pass through the centre

07 Homework

Homework — 4 tasks

This is the last geometry lesson of Grade 10. Keep the year-in-one-page sheet for September.

1. Rewrite in full every annual-paper question that lost a mark, naming the slip in the margin.
2. Write the year in one page: the four quarter ideas with one sketch each.
3. In a cube of edge 10 cm, find the angle between the space diagonal and the base, and the area of the diagonal section.

4. Write one sentence saying what you will keep from this year’s geometry, and one saying what you will practise before September.